

Basic Principles of Medicine 2
Module : Digestion and Metabolism
Lecture No: DM 06

Title: PDH and TCA Cycle

Dr. Mary Maj
Email: mmaj@sgu.edu
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Recommended Reading

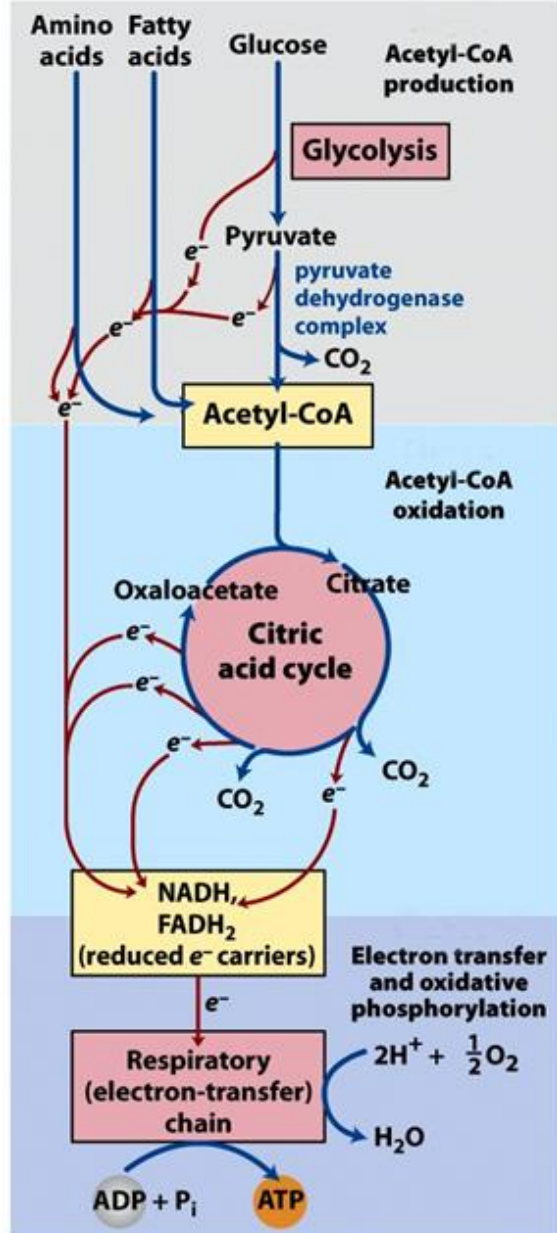
- Lippincott's Biochemistry 9th Edition
 <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
- **Lippincott's Biochemistry:**
 - **Chapter 9.** p120-127:
 - [Tricarboxylic Acid Cycle and Pyruvate Dehydrogenase Complex | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
 - Questions: 9.1 - 9.4
 - Concept map: p.126
- Mark's Basic Medical Biochemistry - 6th Edition, Chapter 23
 - [Tricarboxylic Acid Cycle | Marks' Basic Medical Biochemistry: A Clinical Approach, 6e | Medical Education | Health Library \(lwwhealthlibrary.com\)](#)

Pyruvate Dehydrogenase (PDH) and citric acid cycle:

Objectives

SOM. MK.I.BPM2.2.DM.3.BCHM.1108	Indicate how pyruvate is transported from the cytosol into the mitochondrial matrix
SOM. MK.I.BPM2.2.DM.3.BCHM.1109	Summarize overall reaction catalyzed by PDH complex.
SOM. MK.I.BPM2.2.DM.3.BCHM.1110	Outline the multi-enzyme nature of PDH complex and the cofactors that it requires
SOM. MK.I.BPM2.2.DM.3.BCHM.1111	Indicate inhibitory effects of arsenic on PDH
SOM. MK.I.BPM2.2.DM.3.BCHM.1112	Discuss consequences of thiamine deficiency and its effect on PDH and α KGDH complexes in Wernicke-Korsakoff syndrome and beri-beri.
SOM. MK.I.BPM2.2.DM.3.BCHM.1113	Summarize inherited disorders affecting the PDH complex and their metabolic and clinical consequences.
SOM. MK.I.BPM2.2.DM.3.BCHM.1114	Explain the central role of TCA cycle in metabolism, including its catabolic & anabolic functions (amphibolic role)
SOM. MK.I.BPM2.2.DM.3.BCHM.1115	Outline intermediates and enzymes of the TCA cycle
SOM. MK.I.BPM2.2.DM.3.BCHM.1116	Explain energetics of the cycle, including reactions where reducing equivalents (NADH and FADH ₂) are produced, and the production of GTP by substrate level phosphorylation
SOM. MK.I.BPM2.2.DM.3.BCHM.1117	Explain why aerobic metabolism of glucose using TCA cycle is much more efficient than anaerobic glycolysis
SOM. MK.I.BPM2.2.DM.3.BCHM.1118	Explain regulation of the TCA cycle highlighting the role of NADH, ATP, succinyl-CoA, Ca ²⁺ and oxaloacetate.
SOM. MK.I.BPM2.2.DM.3.BCHM.1119	Indicate the effect of fluorocitrate and malonate on TCA cycle

Catabolism of Proteins, Fats & Carbohydrates: Three stages of Cellular Respiration



1: Oxidation of fatty acids, glucose, and some amino acids yields acetyl-CoA

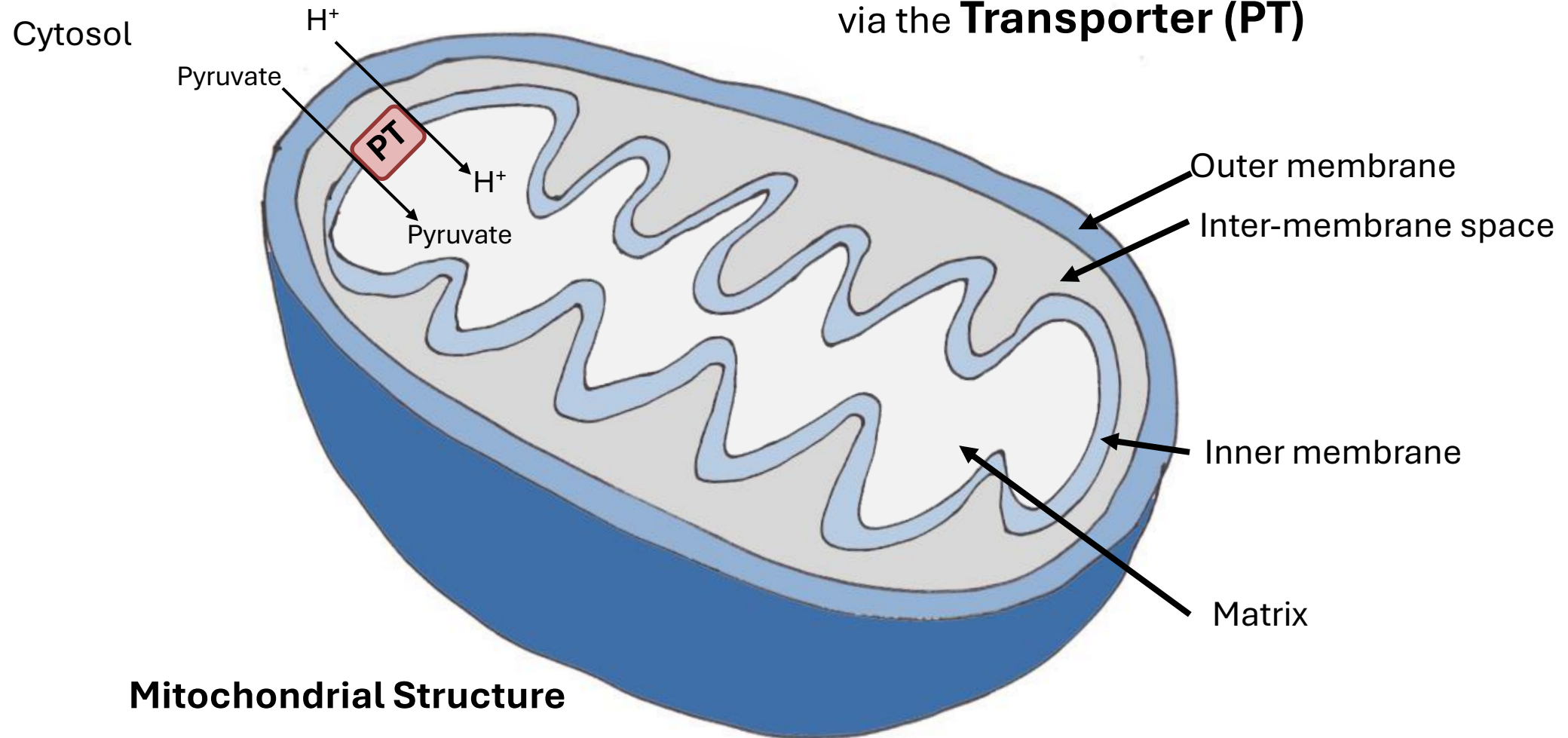
2: Oxidation of acetyl groups in the citric acid cycle includes four steps in which electrons are abstracted

3. Electrons carried by NADH and FADH_2 are funneled into the respiratory chain and the production of ATP.

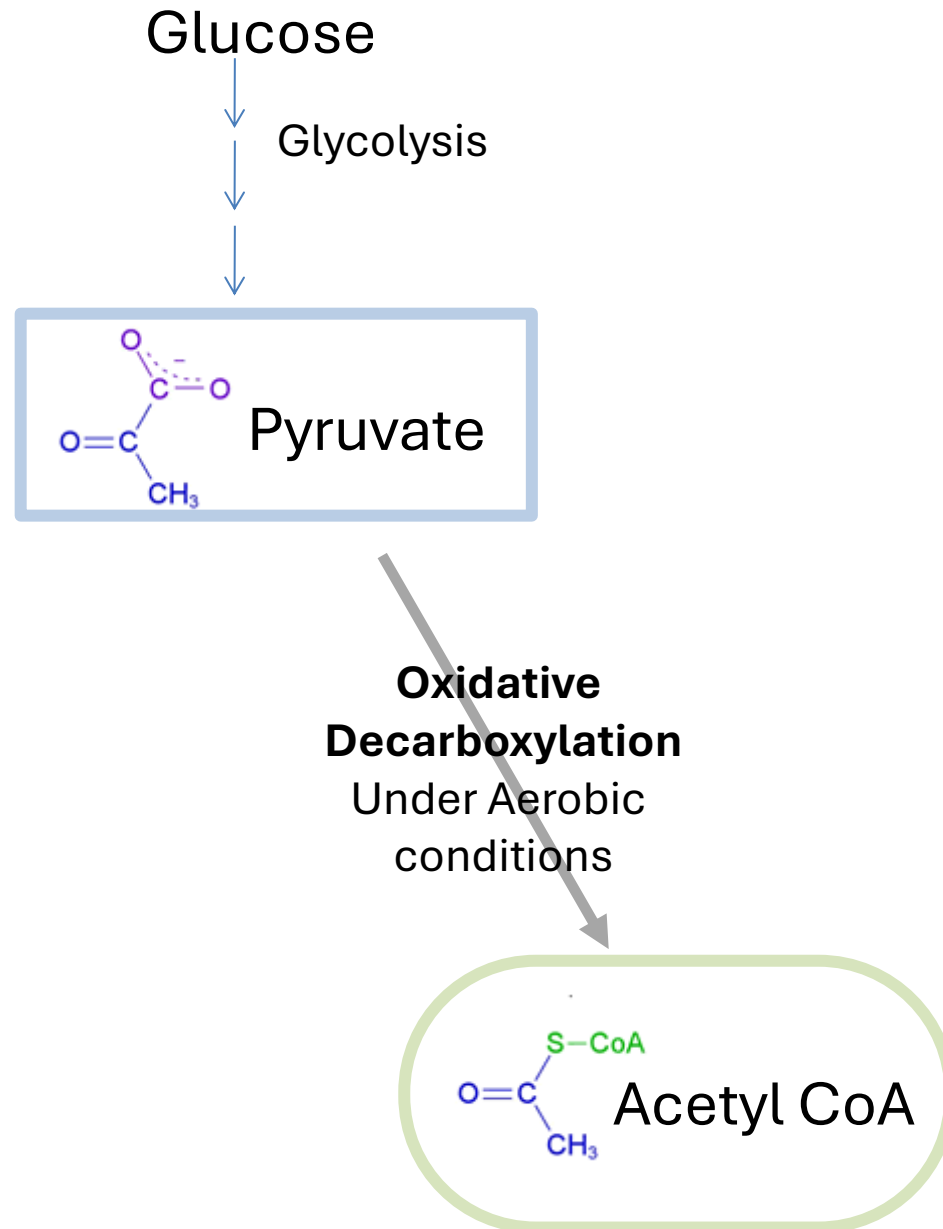
Figure 16-1
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- Problem** - Pyruvate is generated in the cytosol by Glycolysis
- Enzymes for conversion of pyruvate to Acetyl-CoA exist in the mitochondria

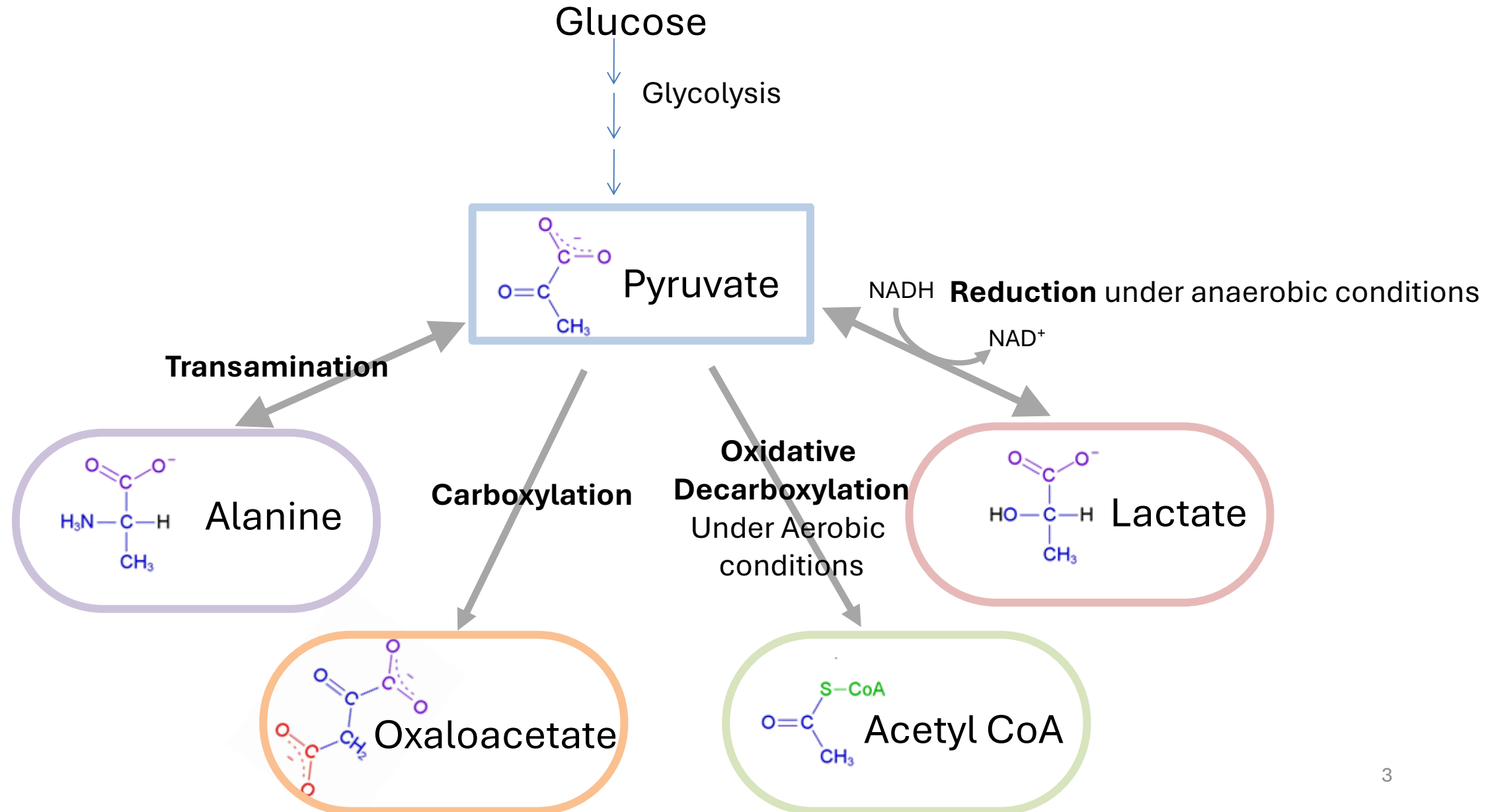
Solution – Pyruvate is transported into the mitochondria via the **Transporter (PT)**



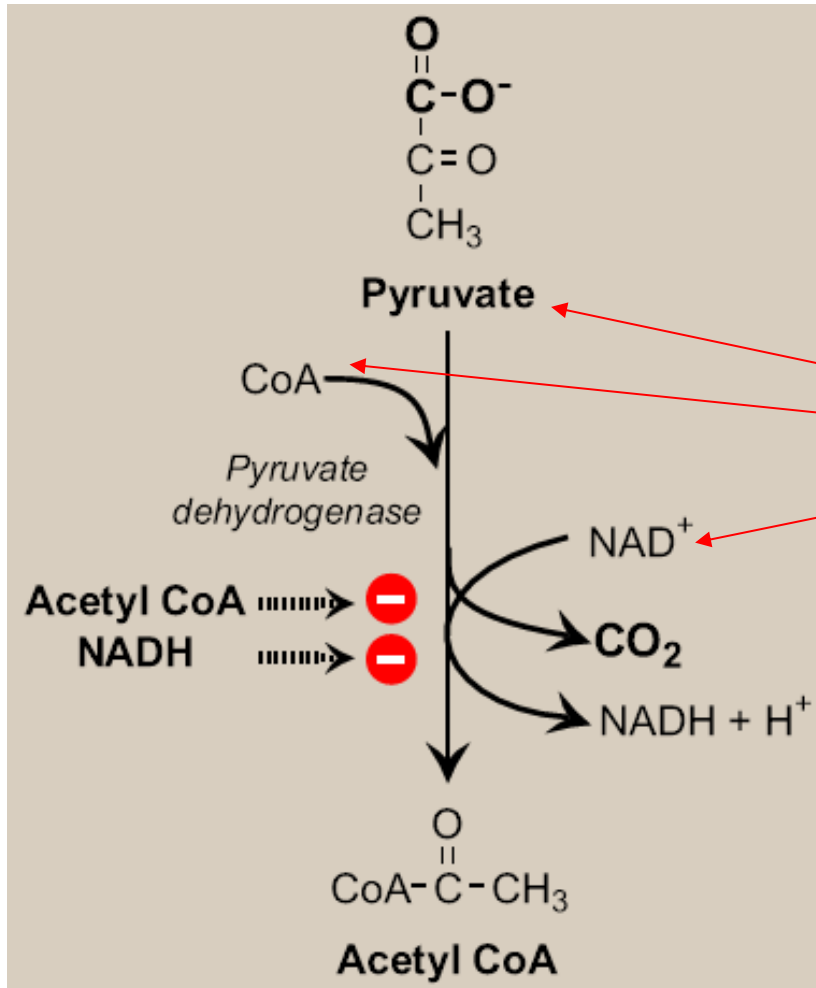
Under Normal Aerobic conditions Pyruvate is Converted to Acetyl CoA



Alternate Fates of Pyruvate



Pyruvate is Oxidized to Acetyl-CoA by the Pyruvate Dehydrogenase Complex



The reaction is termed Oxidative Decarboxylation due to the removal of a carboxyl group and two electrons to generate CO_2 and NADH

The reaction is **irreversible**

Subject to substrate availability

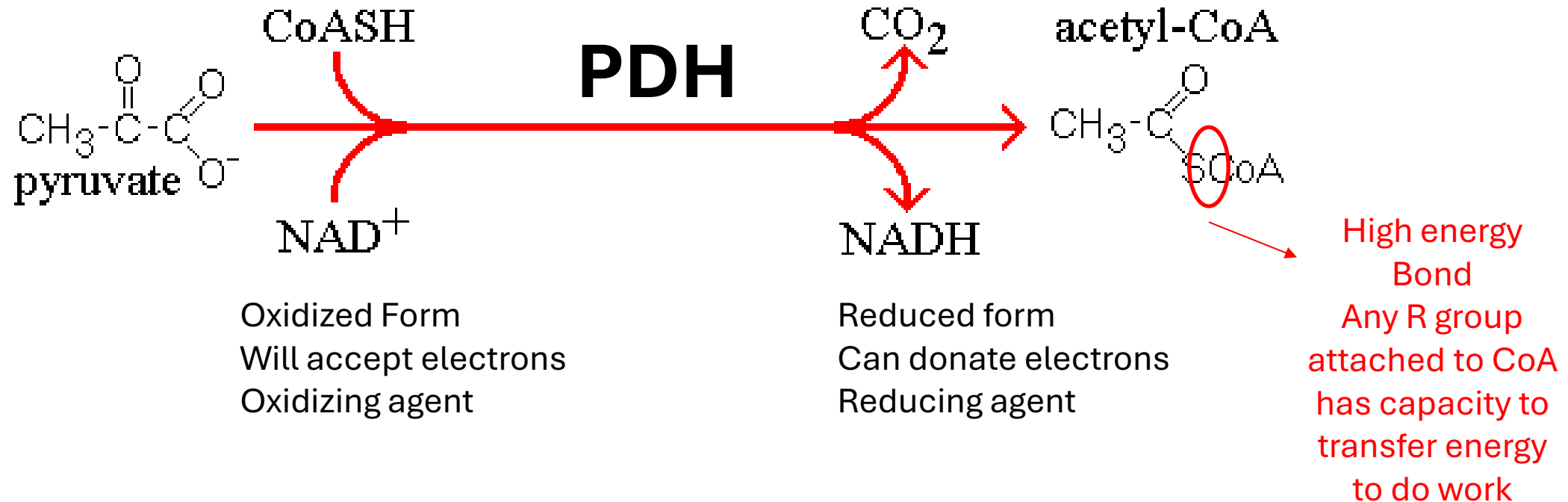
CO_2 is removed by exhalation

NADH can be used to generate ATP via oxidative phosphorylation

Acetyl-CoA enters the Tricarboxylic Acid Cycle (TCA)

Pyruvate Dehydrogenase Complex (PDH)

catalyzes the oxidative decarboxylation of pyruvate



LEO says GER

Losing electrons is oxidation and the substance that loses the electrons is the reducing agent

Gain electrons is reduction and the substance that gains the electrons is the oxidizing agent

PDH is a multienzyme complex with 3 catalytic domains:

- Pyruvate decarboxylase – E1 E1 also called Pyruvate dehydrogenase
- Dihydrolipoyl transacetylase – E2
- Dihydrolipoyl dehydrogenase – E3

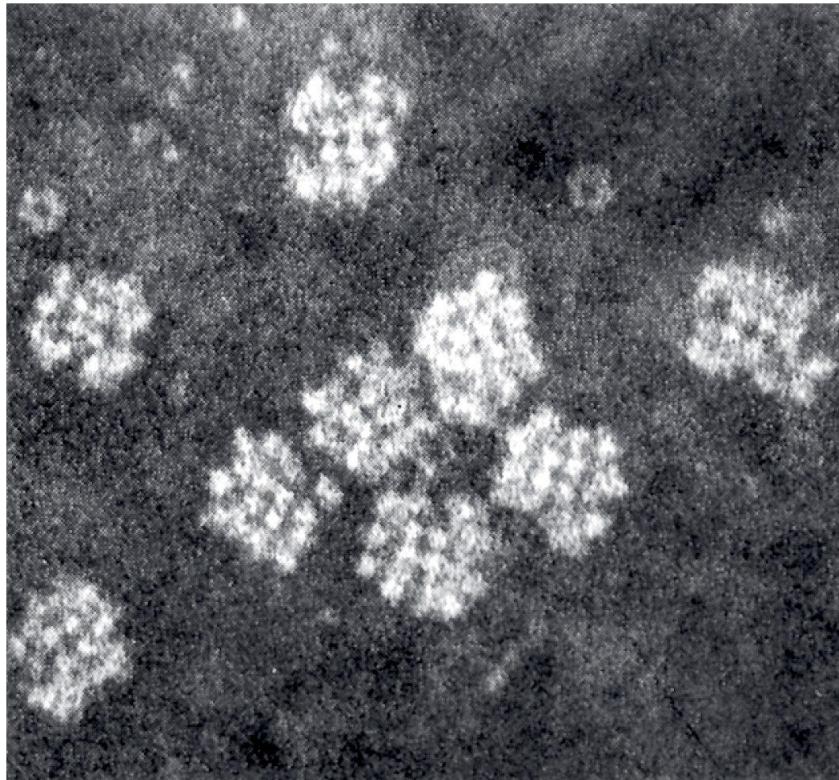
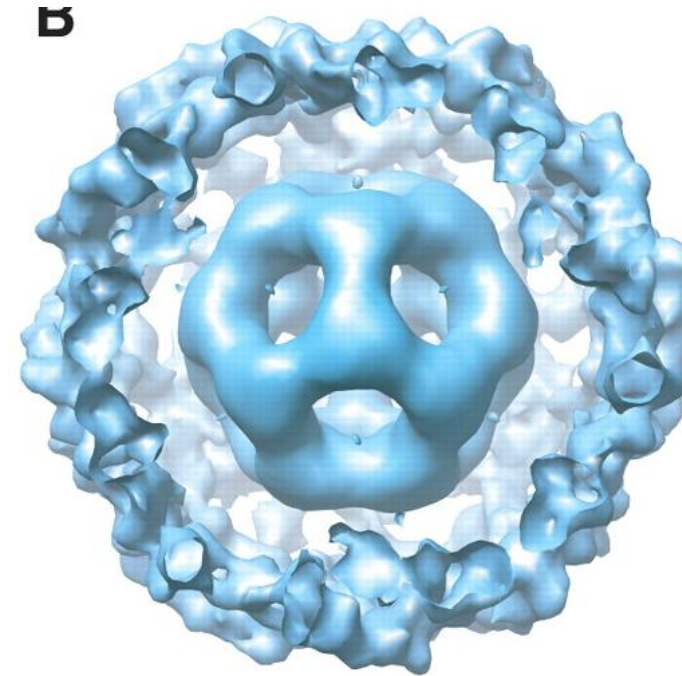


Figure 17-5
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Milne J L S et al. J. Biol. Chem. 2006;281:4364-4370

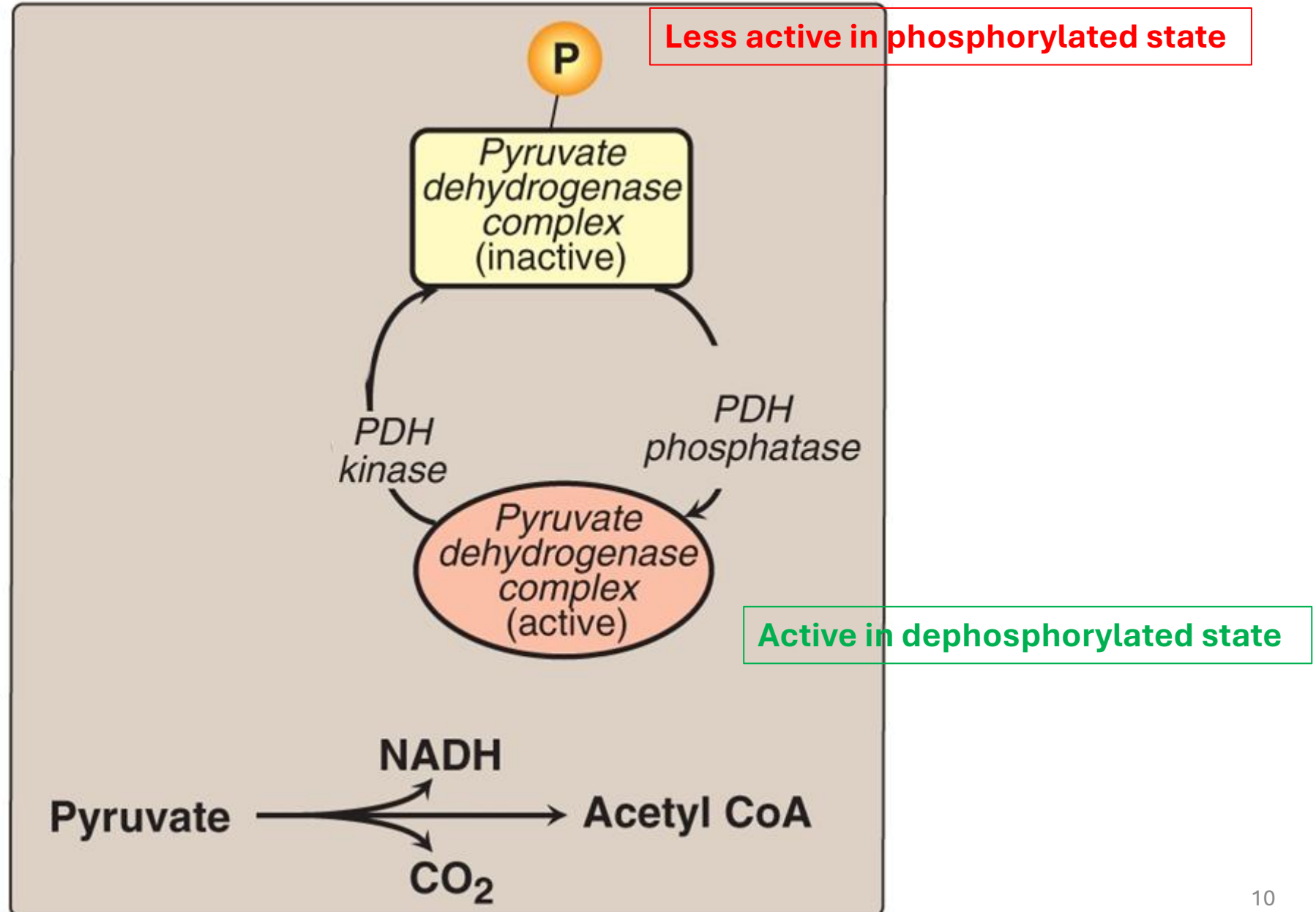
Pyruvate Dehydrogenase Complex (PDH)

- The complex plays a pivotal role in metabolism,
- Limiting the rate of oxidative glucose consumption,
- Highly regulated to respond to all metabolic requirements.
- Reversible phosphorylation of complex by associated regulatory enzymes: PDH kinases and PDH phosphatases
 - PDH Active in dephosphorylated state
 - PDH Less active in phosphorylated state
- **Inhibitors:** (Favor phosphorylation) **NADH, Acetyl-CoA, ATP**
- **Activators:** (Favor dephosphorylation) **Calcium; Pyruvate**

Regulation of PDH

High Energy Charge
OFF

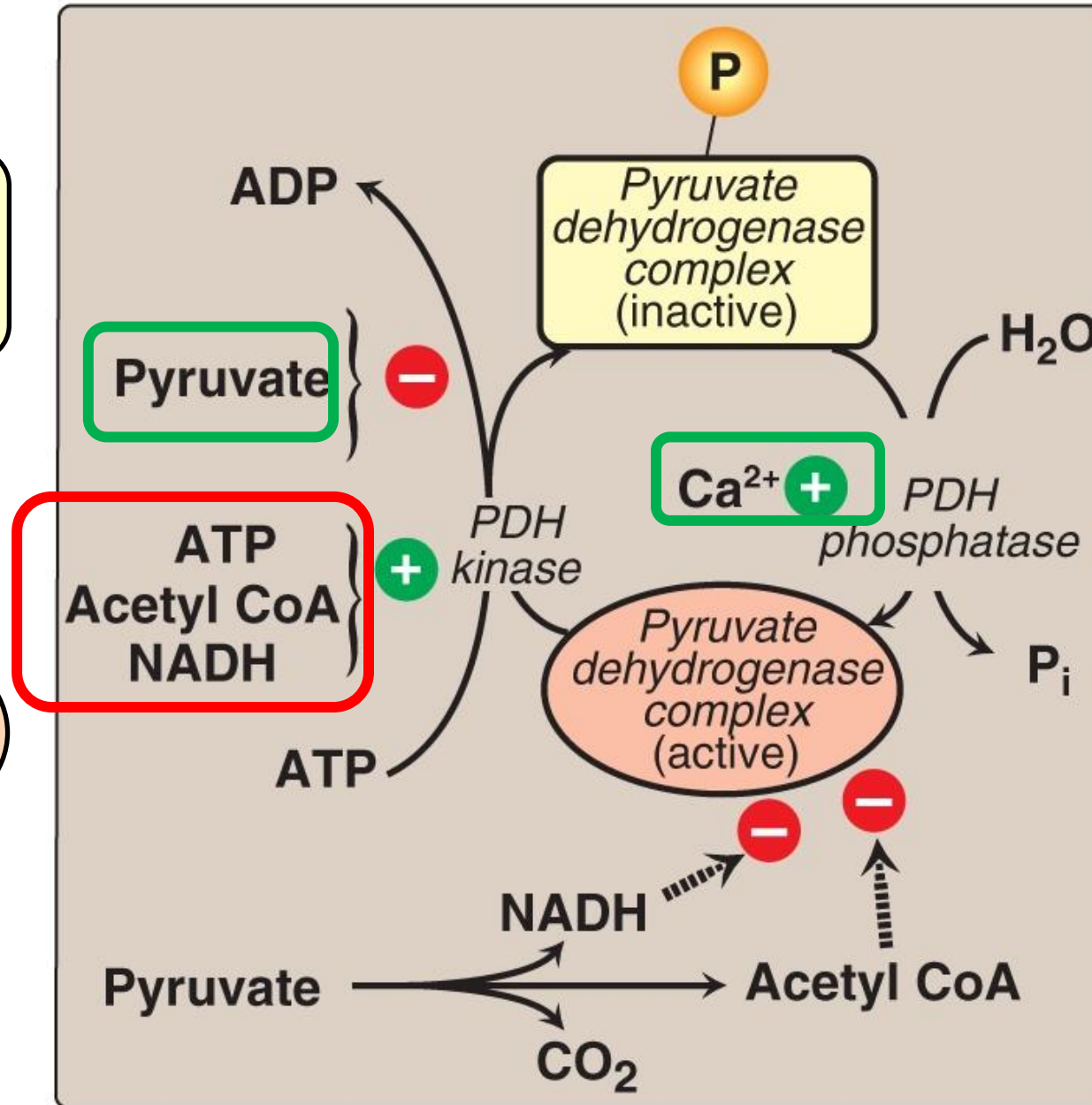
Low Energy Charge
ON



Regulation of PDH

High Energy Charge
OFF

Low Energy Charge
ON



Regulatory molecules:
Inhibitors: (Favor phosphorylation) **NADH, Acetyl-CoA, ATP**
Activators: (Favor dephosphorylation) **Calcium; Pyruvate**

Skeletal Muscle
Ca²⁺ release during contraction → Active PDH → Energy production.

Cardiac Muscle
 Catecholamines (stress)

Activation of PDH Complex

Active in dephosphorylated state

Dephosphorylated state:

- **Calcium**, particularly in skeletal muscle during contraction
 - **Ca²⁺ release** during contraction → Active PDH → Energy production.
- Pyruvate (substrate of PDH)
- **Insulin** in adipocytes and liver
- Catecholamines (stress) in cardiac muscle

Inhibition of PDH Complex

- Inhibition of PDH by reaction products
 - Acetyl CoA & NADH
- **Inhibition of PDH by phosphorylation**
 - ATP, Acetyl CoA and NADH

Pyruvate Dehydrogenase Complex

- **Bound prosthetic groups**
 - Thiamine pyrophosphate (TPP) – from B1
 - Lipoic acid – from octanoic acid
 - FAD – from B2 (riboflavin)
- **Cofactors**
 - NAD^+ - from B3 (Niacin)
 - Coenzyme A – from B5 (Pantothenic acid)

The 5 cofactors

Co-factors for PDH and Alpha Ketoglutarate:
"Tender Loving care for Nancy"

TPP

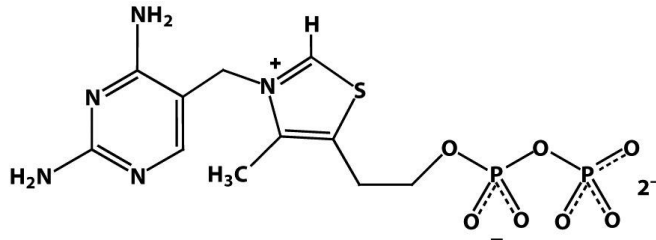
Lipoic acid

Coenzyme A

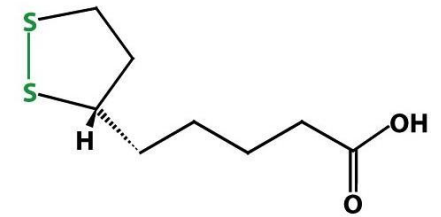
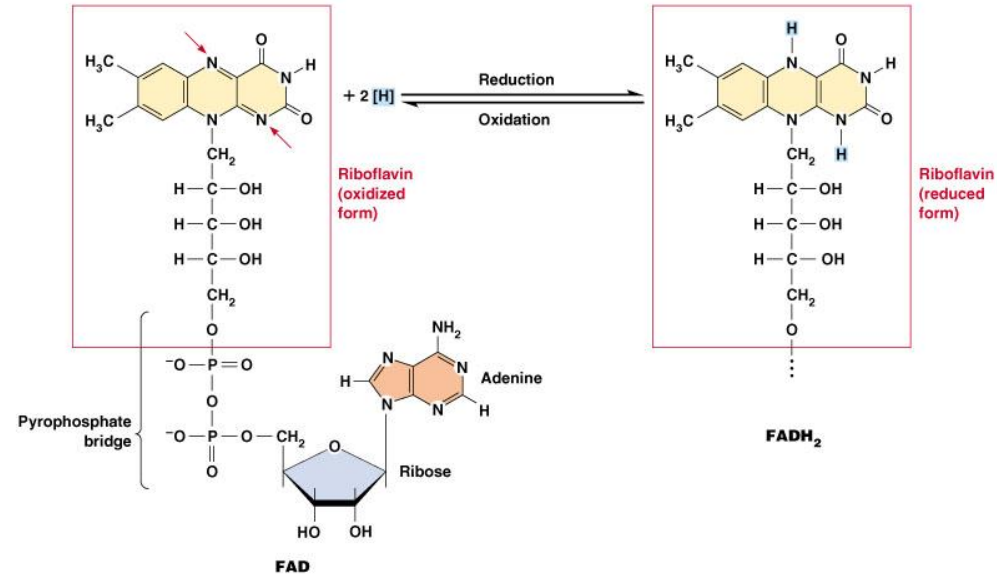
FAD

NAD

Bound Prosthetic groups

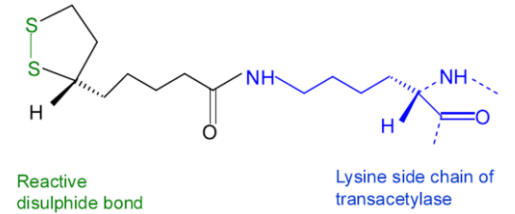


Thiamine pyrophosphate (TPP)

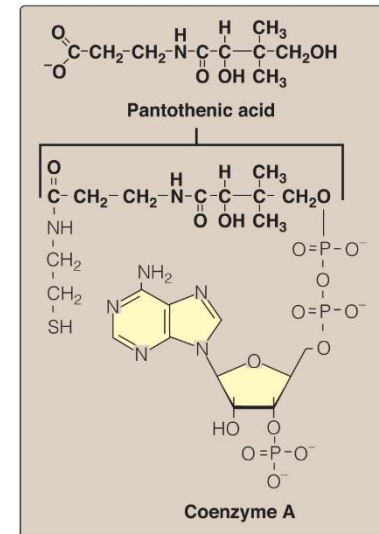
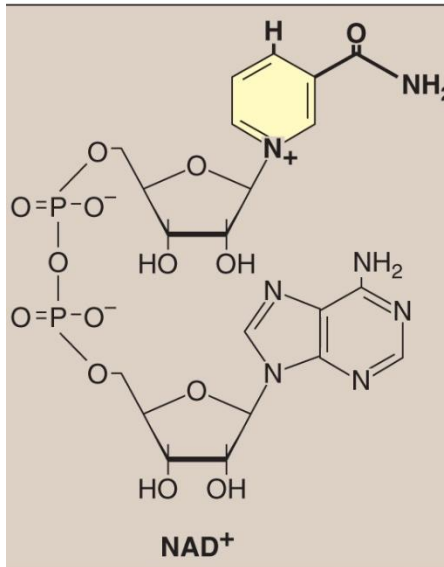


Lipoic acid

Lipoamide

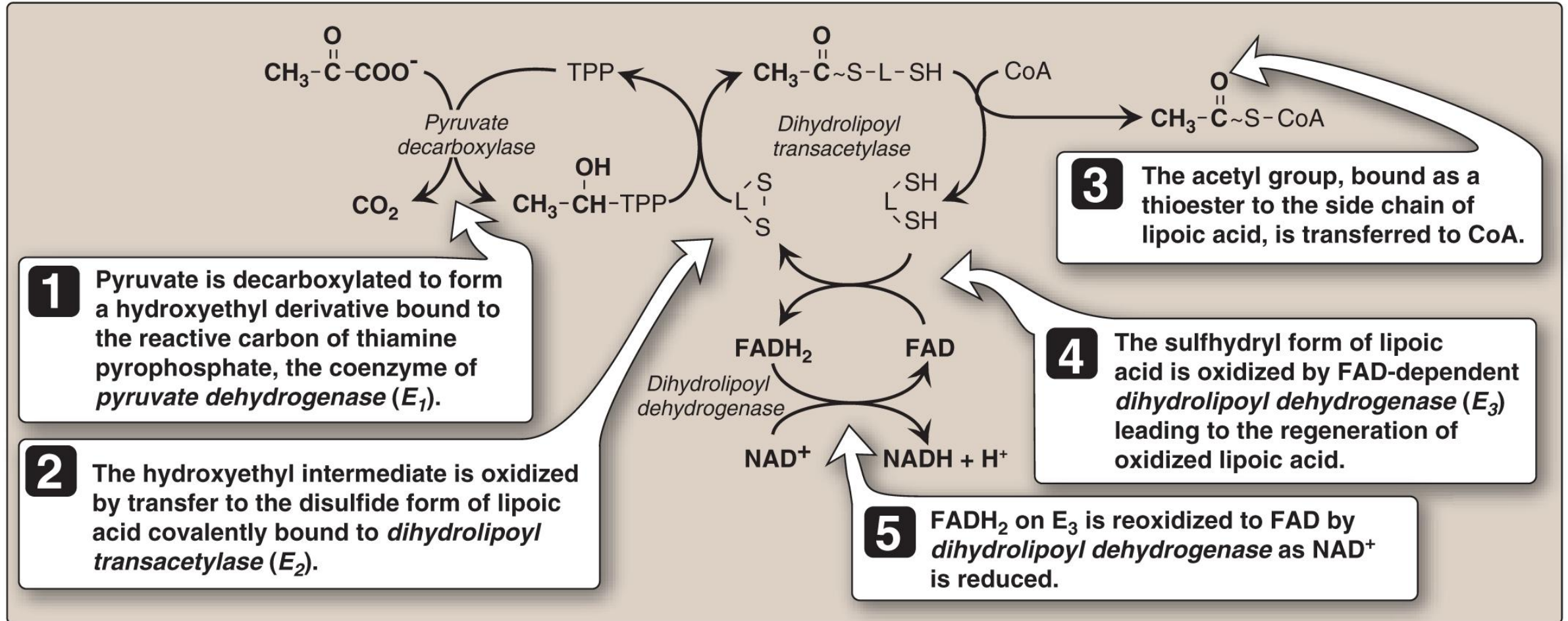


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Pyruvate Dehydrogenase Complex



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Intricacies of the PDH complex: Do NOT memorize

NADH to the electron transport chain

Clinical Notes: PDH Deficiencies

- **Several vitamins required for PDH activity:**
 - Vit B₅ (Pantothenic acid) → Co-enzyme A,
 - Vit B₃ (Niacin) → NAD,
 - Vit B₂ (Riboflavin) → FAD,
 - Vit B₁ (Thiamine) → TPP
 - Deficiency in any of these vitamins will disrupt PDH activity

Clinical Notes: Thiamine Deficiency Causing Low PDH Activity

- **Wernicke-Korsakoff syndrome** characterized by Ataxia, Ophthalmoplegia, Memory loss, Cerebral Hemorrhage, and Confabulation
 - At risk: Alcoholics & malnourished individuals
- **Beriberi**: wet and dry based on presence/absence of edema
 - Wet Beriberi: Heart failure, decreased ATP, increased cardiac output, (dilated cardiomyopathy)
 - Dry Beriberi: systemic muscle wasting, polyneuropathy

Other thiamine-requiring enzymes:

- α -ketoglutarate DH (α KG DH) part of the TCA cycle
- Branched chain α -ketoacid DH (BCKDH)
- Transketolase (Pentose phosphate pathway)

Clinical notes: Pyruvate dehydrogenase deficiency

Metabolic effect:

↑ in pyruvate with concomitant ↑ in lactic acid and alanine (via transamination), ↓ in production of acetyl CoA; severe reduction in ATP production

Clinical features:

lactic acidosis in blood and CSF, increased blood pyruvate and alanine, neurologic defects, myopathy; usually fatal at early age

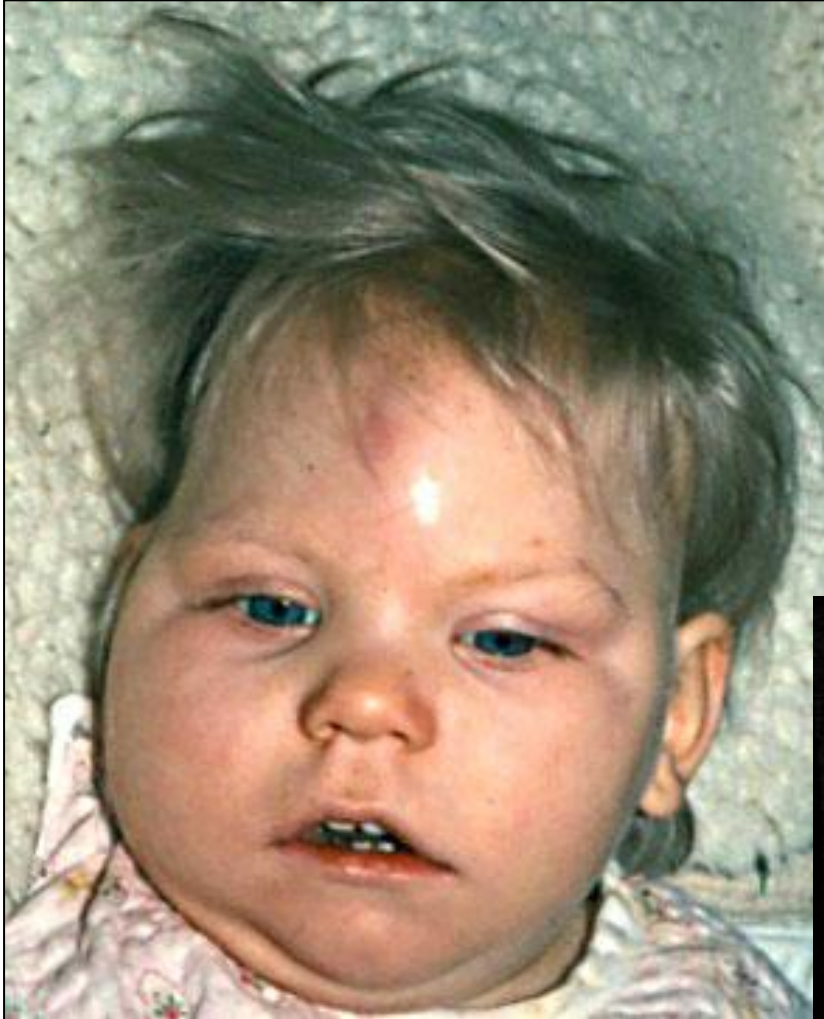
Therapies:

Dichloroacetate inhibits PDH kinase so that PDH complex remains unphosphorylated and catalytically as active as possible

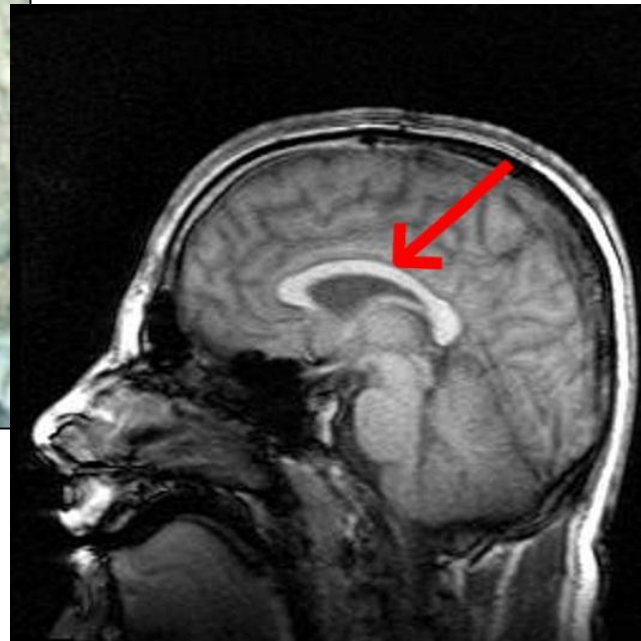
Supplementation with **thiamine**, **lipoic acid** and carnitine

High fat and low carbohydrate diet

PDH Deficiency

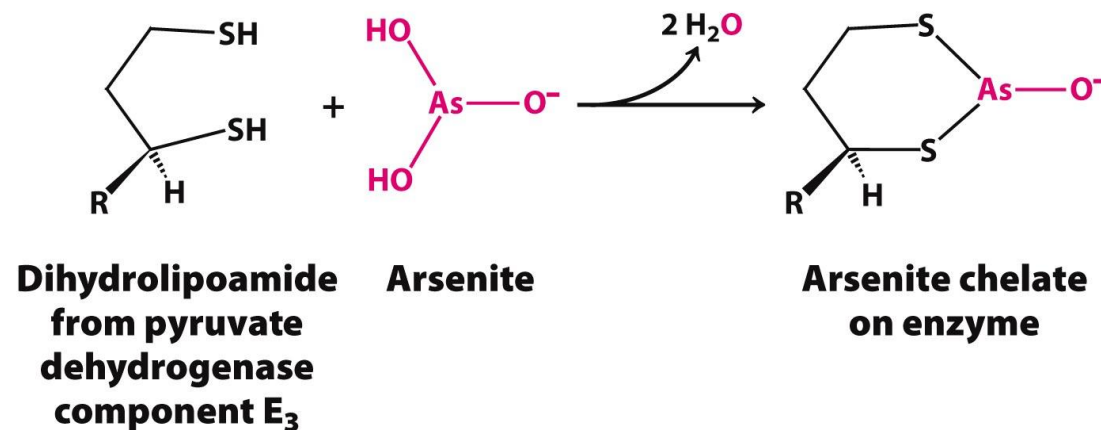


- Frontal prominence
- Wide nasal bridge
- Flared nares
- Long philtrum
- Brain malformations
 - **CNS solely depends on Glucose Metabolism**
 - Corpus callosum agenesis
 - Cerebral and basal ganglia cysts



Heavy metal poisoning disrupts pyruvate metabolism

- Arsenate, Mercury, and lead have high affinity for -SH
- **Lipoic acid** is one of the cofactors in PDC
- PDH Complex becomes inactive when lipoic acid is bound to heavy metals.
- CNS solely depends on Glucose Metabolism through to oxidative phosphorylation for energy, (can not use fatty acids as a fuel source) therefore affected by heavy metal poisoning.



TCA cycle



Hans Krebs, 1900–1981

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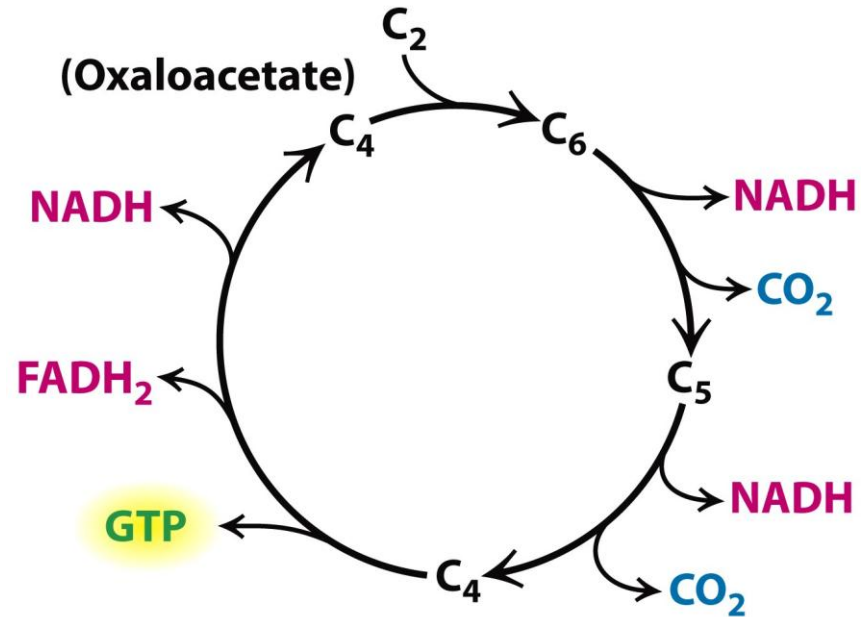


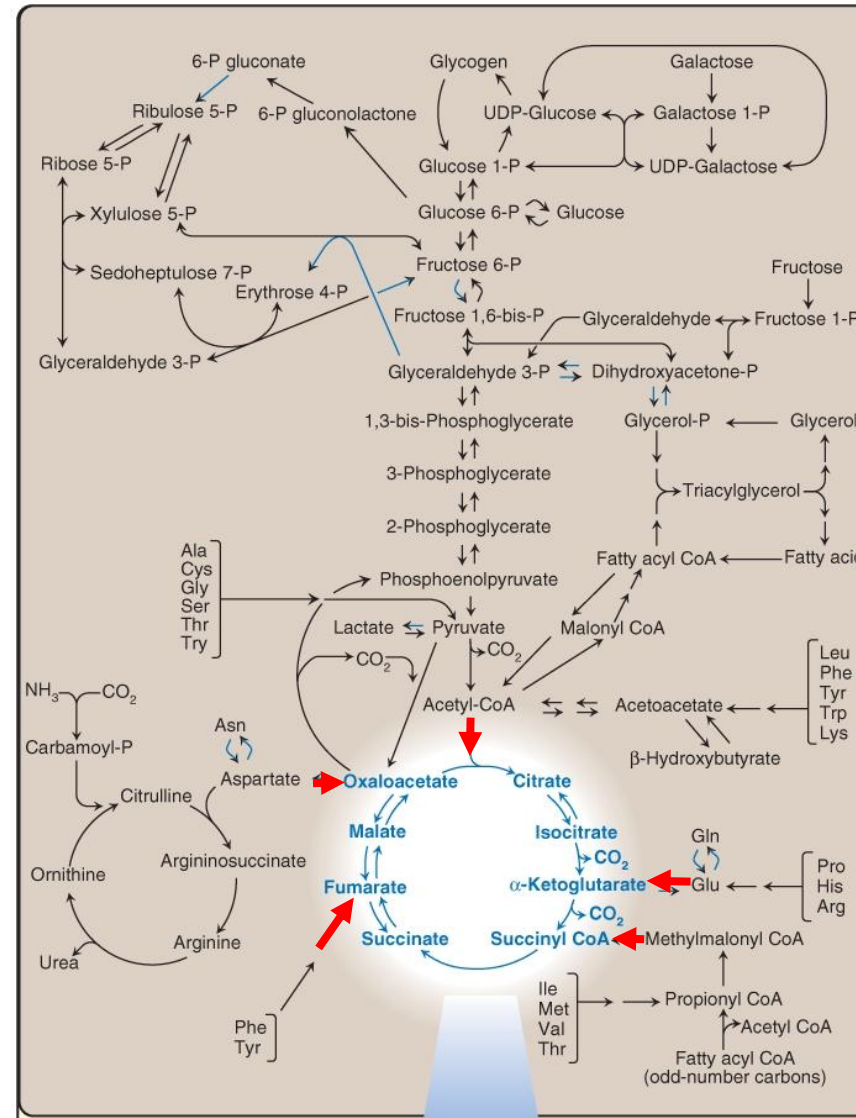
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Tricarboxylic Acid Cycle

Krebs Cycle

Citric Acid Cycle

Input to the TCA cycle



Called Anaplerotic reactions

TCA COMPONENTS: IMPORTANT BIOSYNTHETIC INTERMEDIATES

- **Amphibolic pathway, serves in both catabolic and anabolic processes**
 - Provides precursors for many biosynthetic pathways
 - It is necessary to replace the ones used in biosynthesis of molecules
- **Reactions which replenish TCA acid cycle intermediates are called anaplerotic reactions** (to fill up)
 - Dynamic balance between:
 - Reactions where intermediates are consumed
 - Intermediates which are replenished by the anaplerotic reactions.
 - **So that the concentrations of the citric acid cycle intermediates remain almost constant**

Reactions of TCA cycle

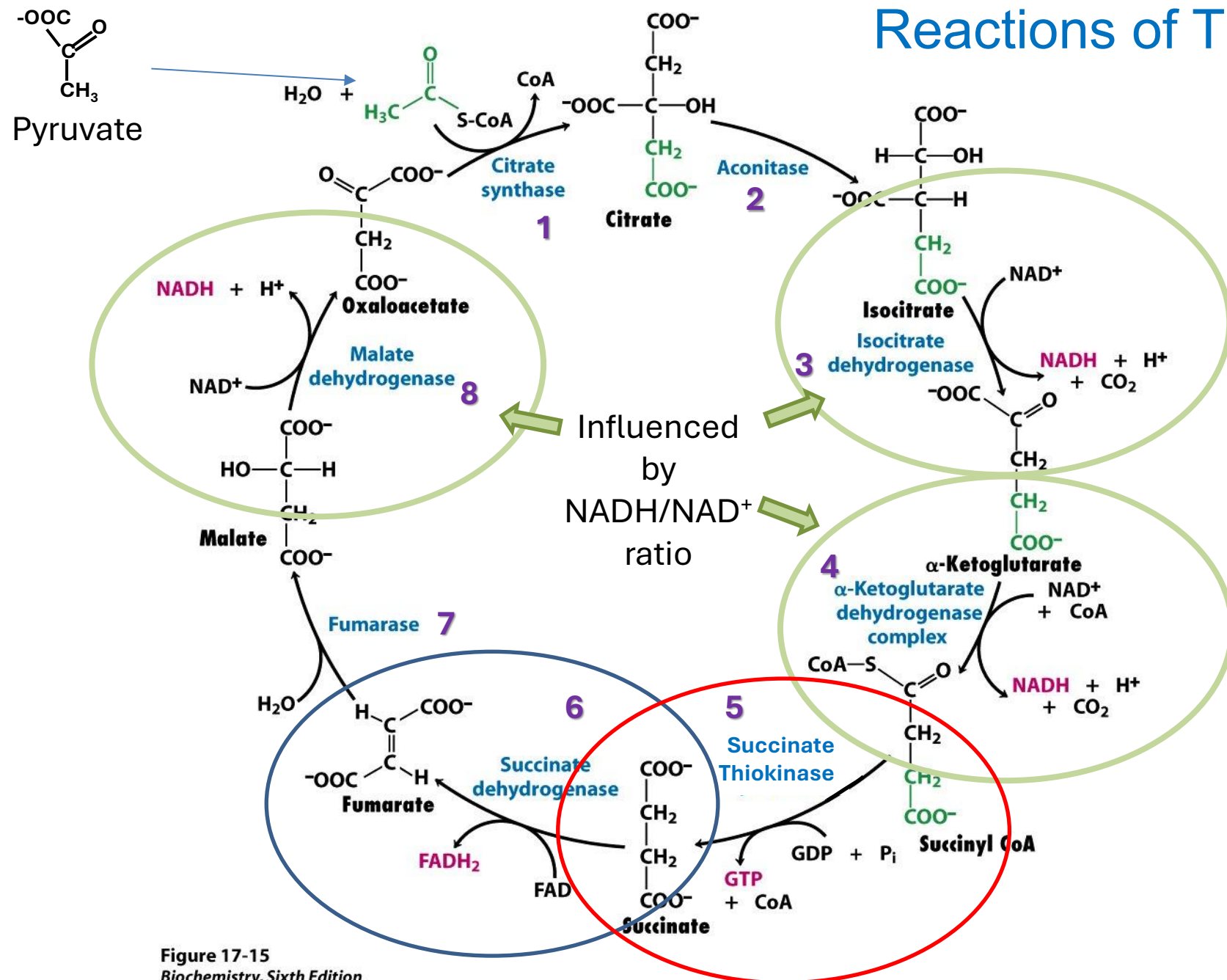
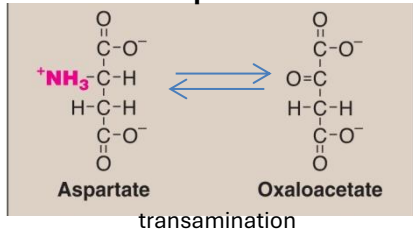
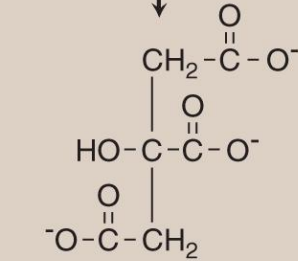
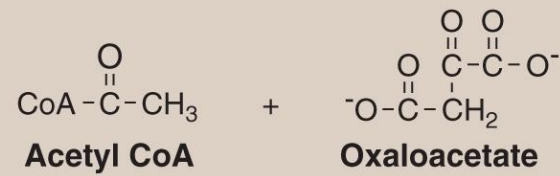


Figure 17-15
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Anaplerotic



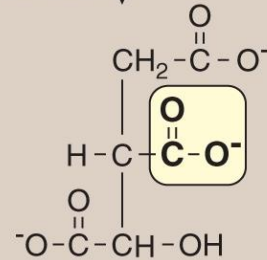
1



Citrate

2

Aconitase

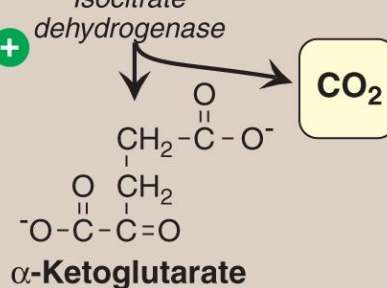


Isocitrate

ATP } -
NADH }



ADP } +
Ca²⁺ }



3

Reactions of TCA cycle

Condensation of acetyl CoA and oxaloacetate to form citrate (tricarboxylic acid) by **Citrate synthase**

Regulated enzyme

(High citrate inhibits citrate synthase in the mitochondria and phosphofructokinase-1 in the cytosol)

Citrate is isomerized to isocitrate by

Aconitase ★

Irreversible oxidative decarboxylation of isocitrate by

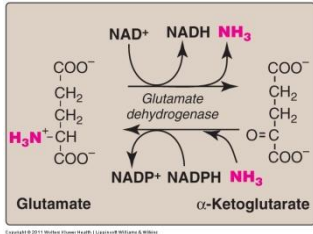
Isocitrate dehydrogenase

(Regulated enzyme)

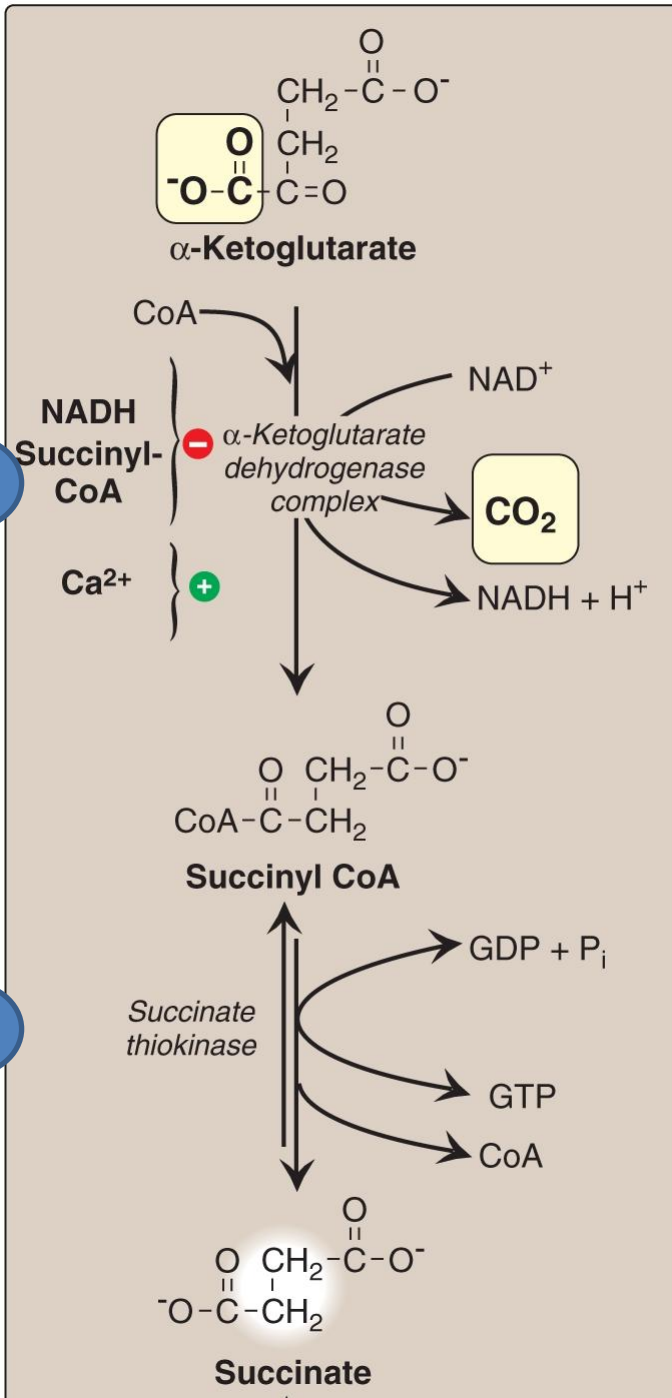
Yields 1 NADH (1 of 3)

Allosteric activation

Anaplerotic



4



5

Reactions of TCA cycle

Conversion of α-ketoglutarate to succinyl CoA by
α-ketoglutarate dehydrogenase (requires TPP, NAD⁺, FAD, Lipoic acid and CoA)
Regulated enzyme

Releases second CO₂ and forms second NADH (2 of 3)

Succinate thiokinase (succinyl CoA synthetase)
 Cleaves high energy thioester bond of succinyl CoA

Yields 1 GTP

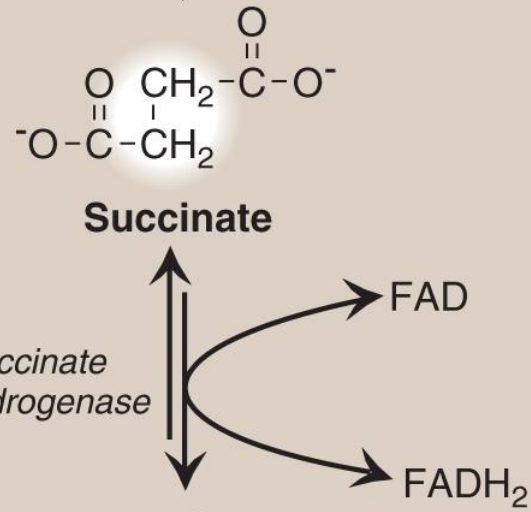
What type of reaction is this???

Substrate-level phosphorylation

Reactions of TCA cycle

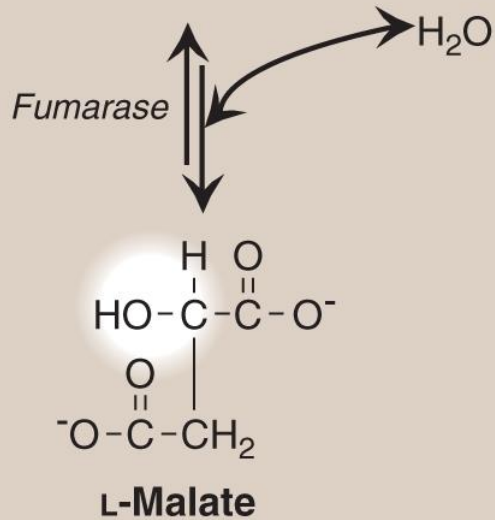
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Succinate dehydrogenase



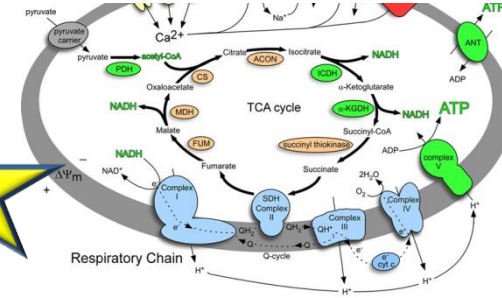
7

Fumarase

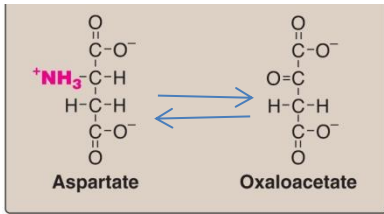


Succinate is oxidized to fumarate by **succinate dehydrogenase**

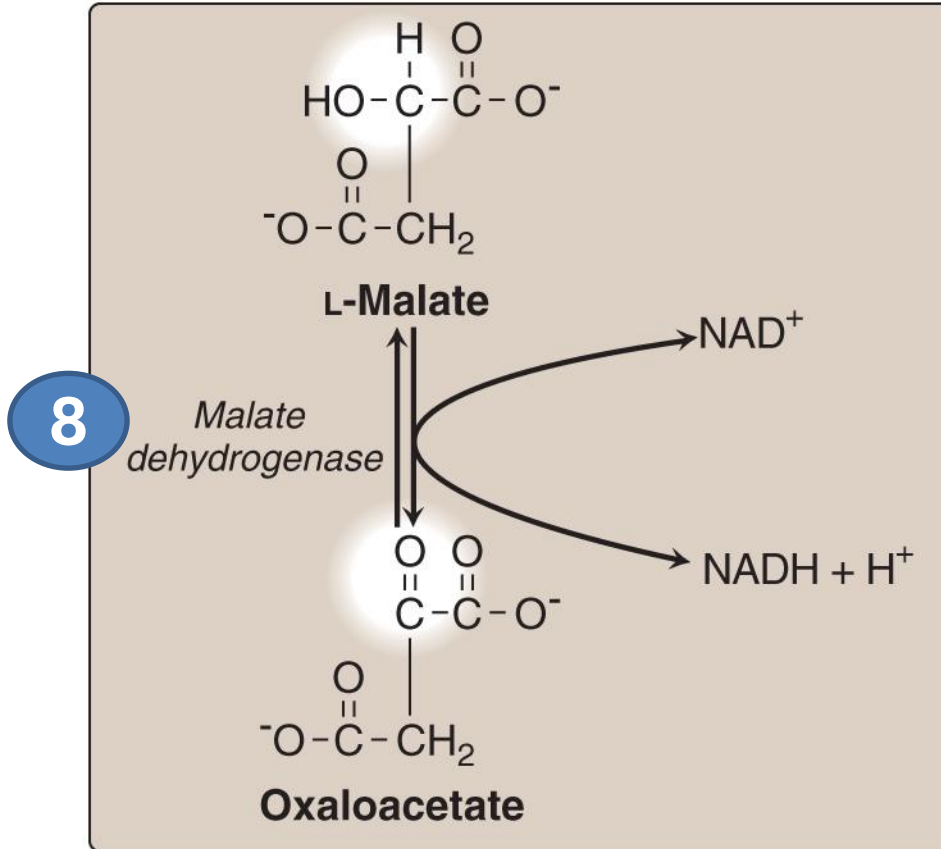
FAD is reduced to FADH₂



Anaplerotic



transamination



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Reactions of TCA cycle

Malate is oxidized to oxaloacetate by **malate dehydrogenase**

Regulated by substrate concentration: Malate and NAD^+

Third NADH formed (3 of 3)

High $\text{NADH} / \text{NAD}^+$ ratio favors **malate formation (Ethanol)**

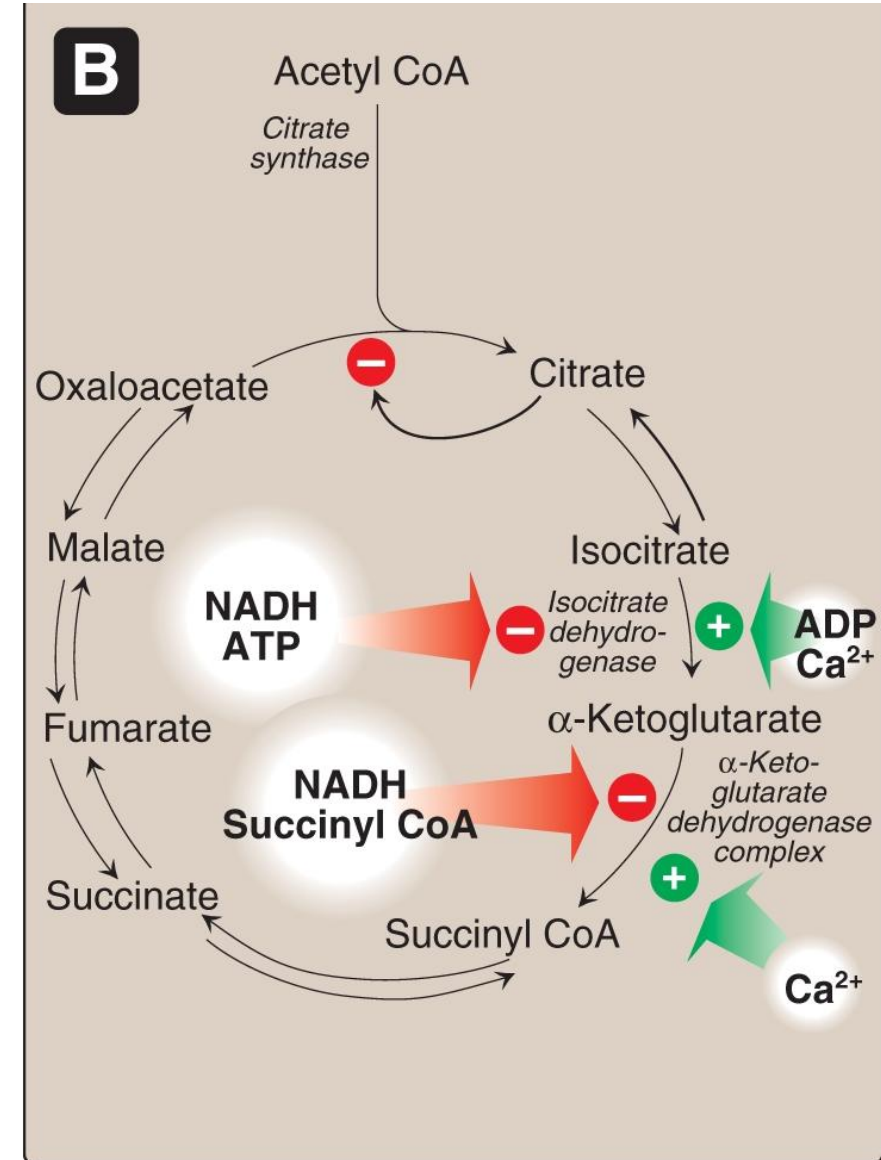
Regulation of TCA:

- Citrate Synthase
- Isocitrate Dehydrogenase
- α -KG Dehydrogenase

High NADH/NAD⁺ ratio $\rightarrow$ Inhibition of TCA cycle in liver

Observed in:

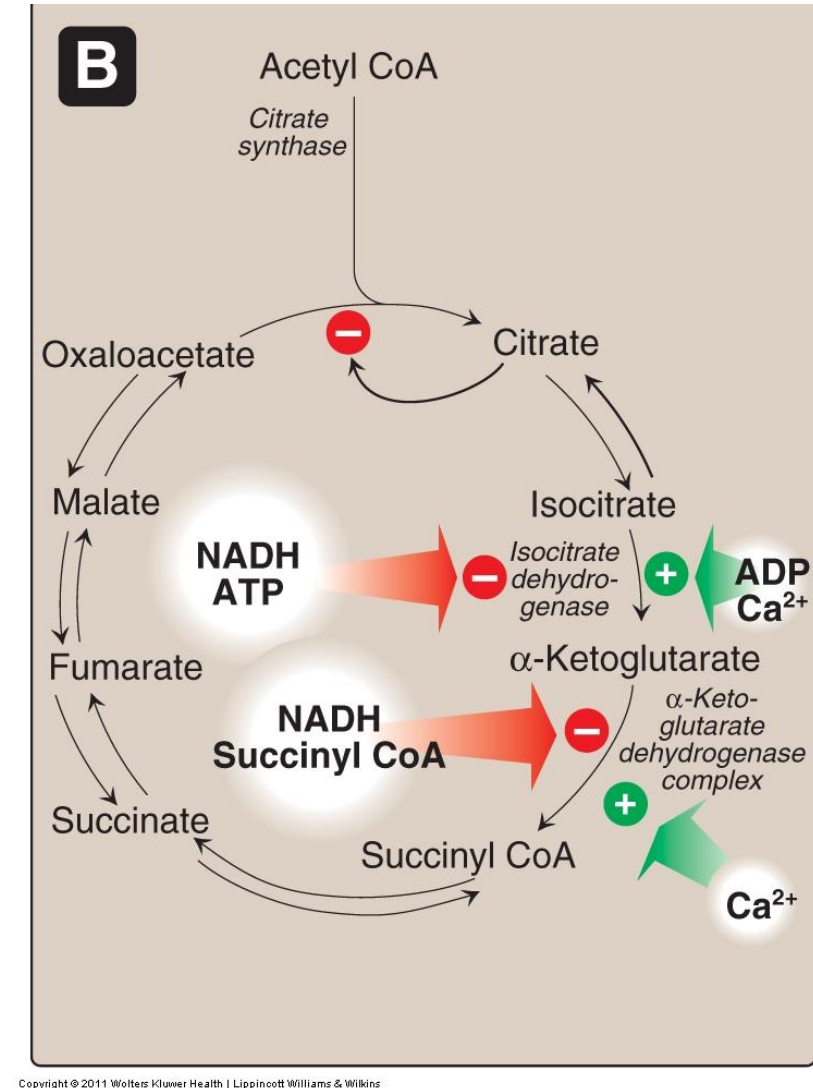
- Fasting (beta-oxidation produces NADH)
- Alcohol binge drinking
- Hypoxia



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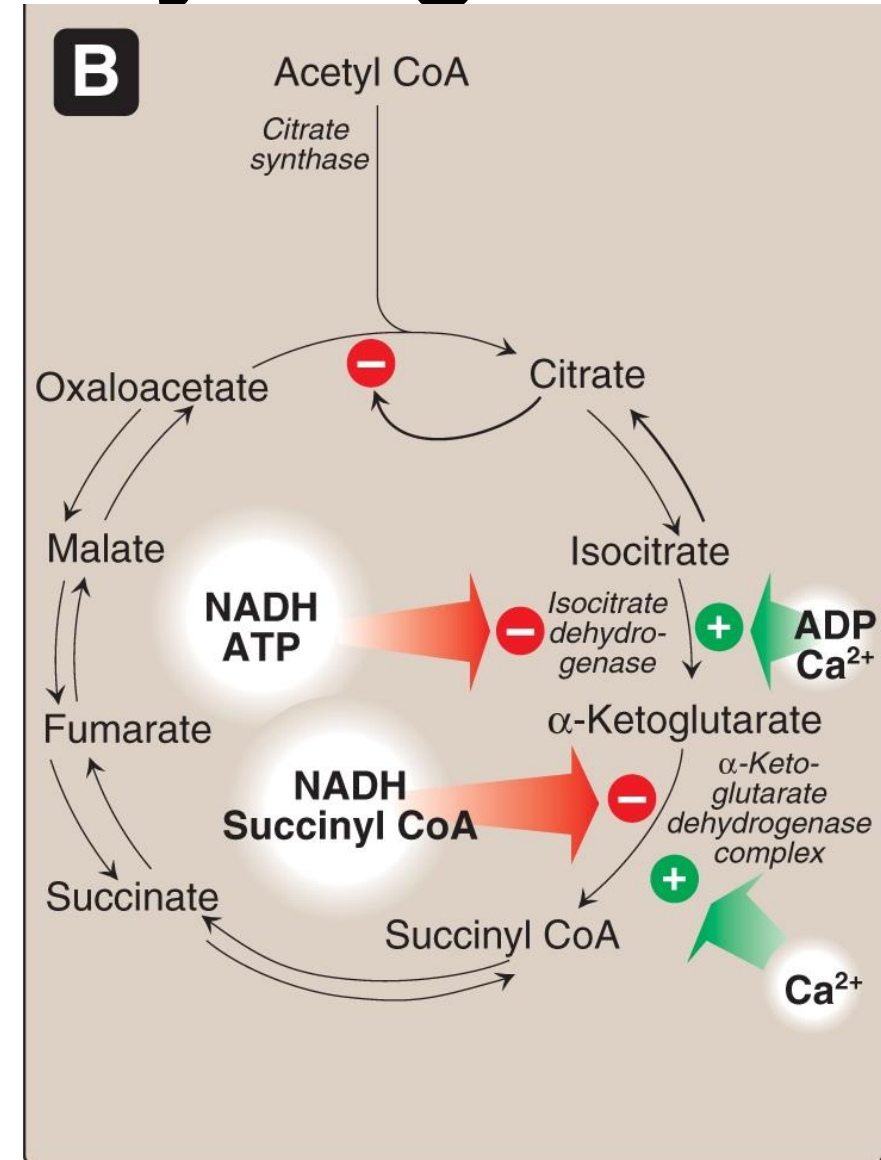
1. Regulation of Citrate Synthase

- irreversible reaction
- inhibited by its product (Citrate)
- Activated primarily by substrate availability (Acetyl CoA and oxaloacetate)



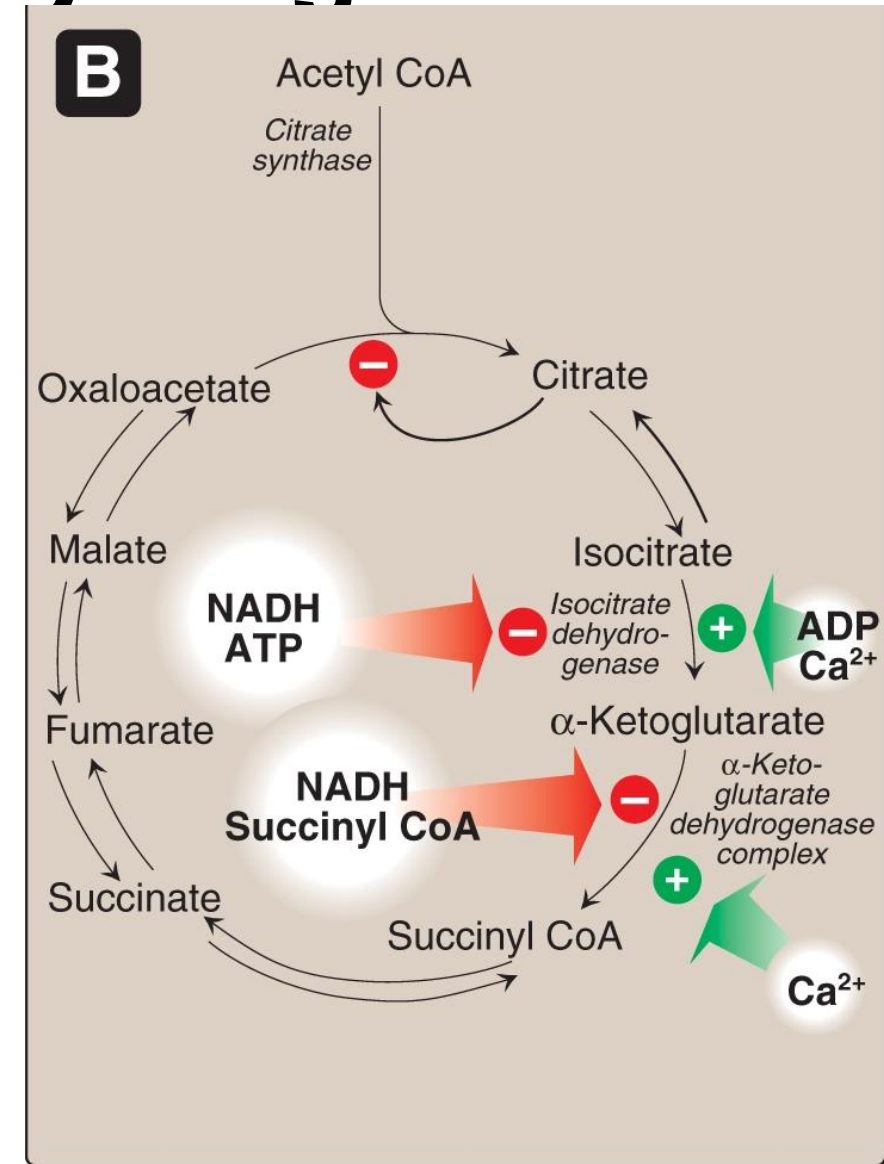
2. Regulation of Isocitrate Dehydrogenase

- Irreversible reaction
- Allosterically activated by ADP & calcium
- Inhibited by ATP, NADH



3. Regulation of α -KG Dehydrogenase

- Structure similar to PDH with E1, E2 & E3
- 5 co-enzymes required, TPP, lipoic acid, FAD, NAD^+ and CoA
- Activated by Ca^{2+}
- Inhibited by NADH and succinyl CoA





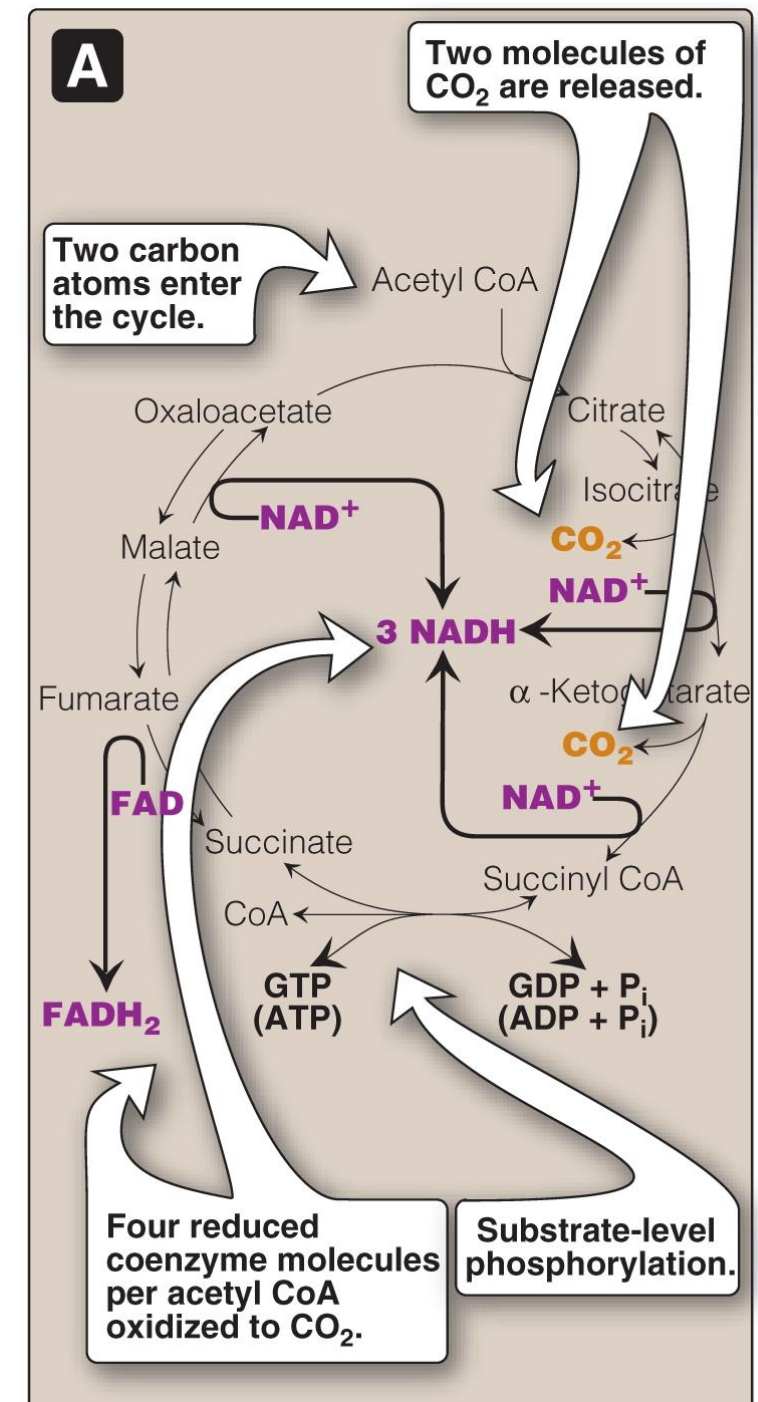
One TCA cycle revolution produces:

3 NADH (9 ATP)

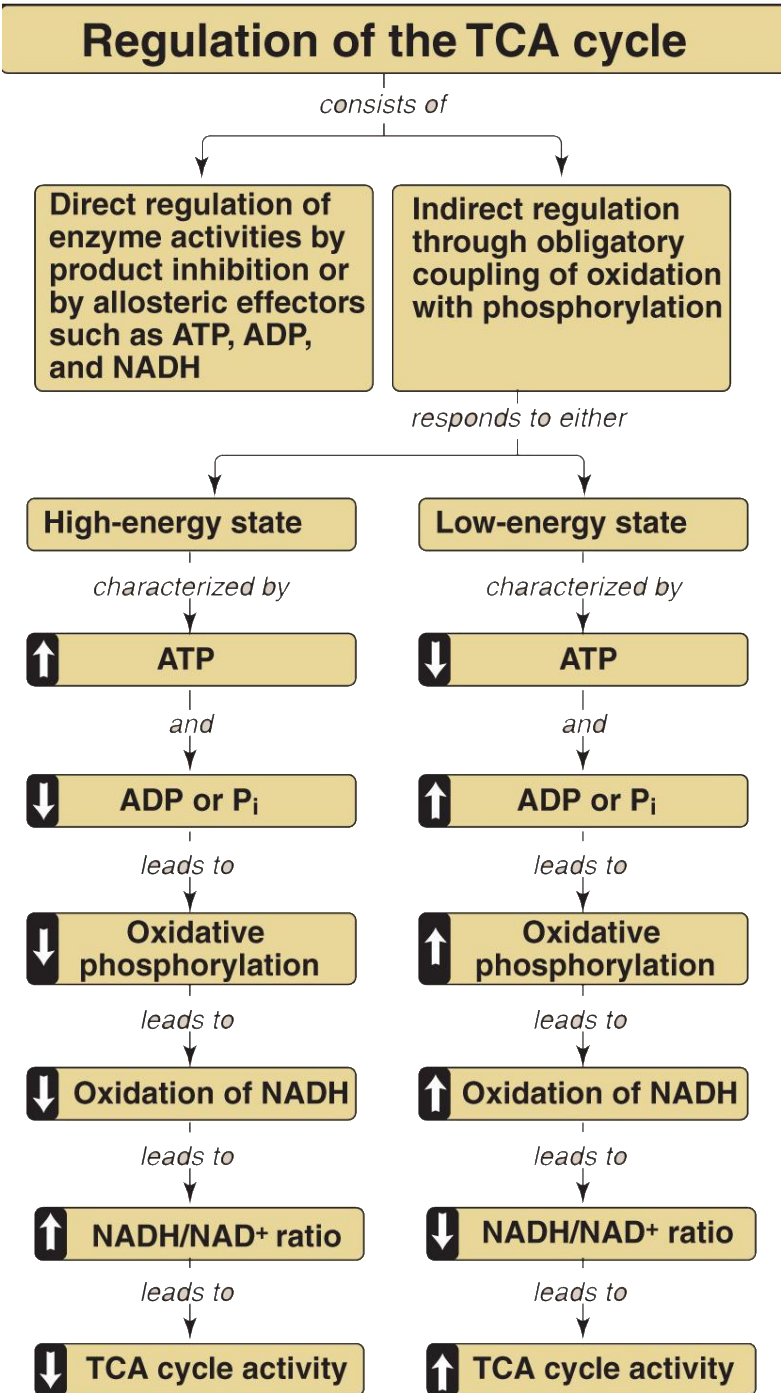
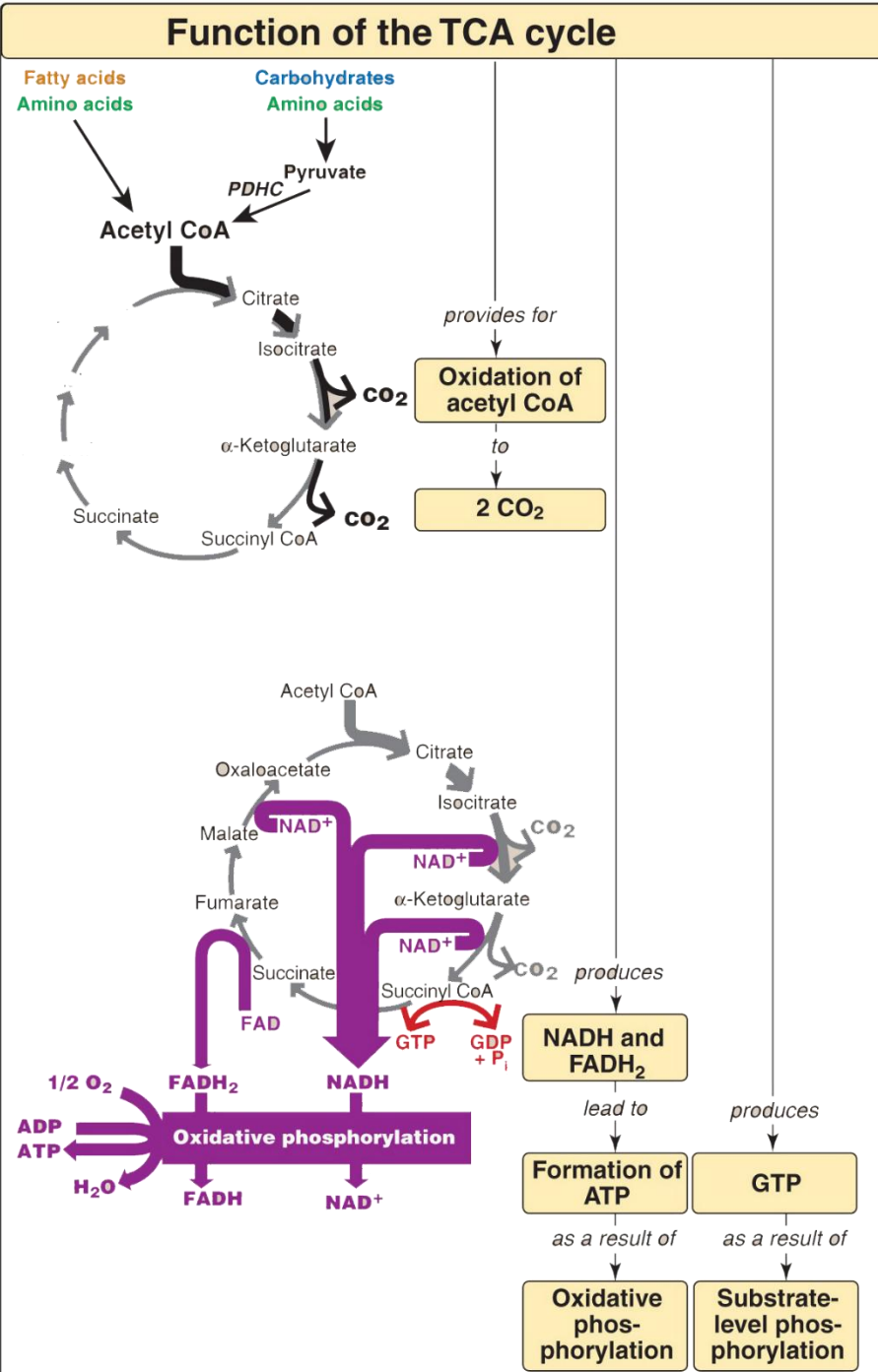
1 GTP (1 ATP)

1 FADH₂ (2 ATP)

12 ATP per acetyl CoA oxidized



Summary



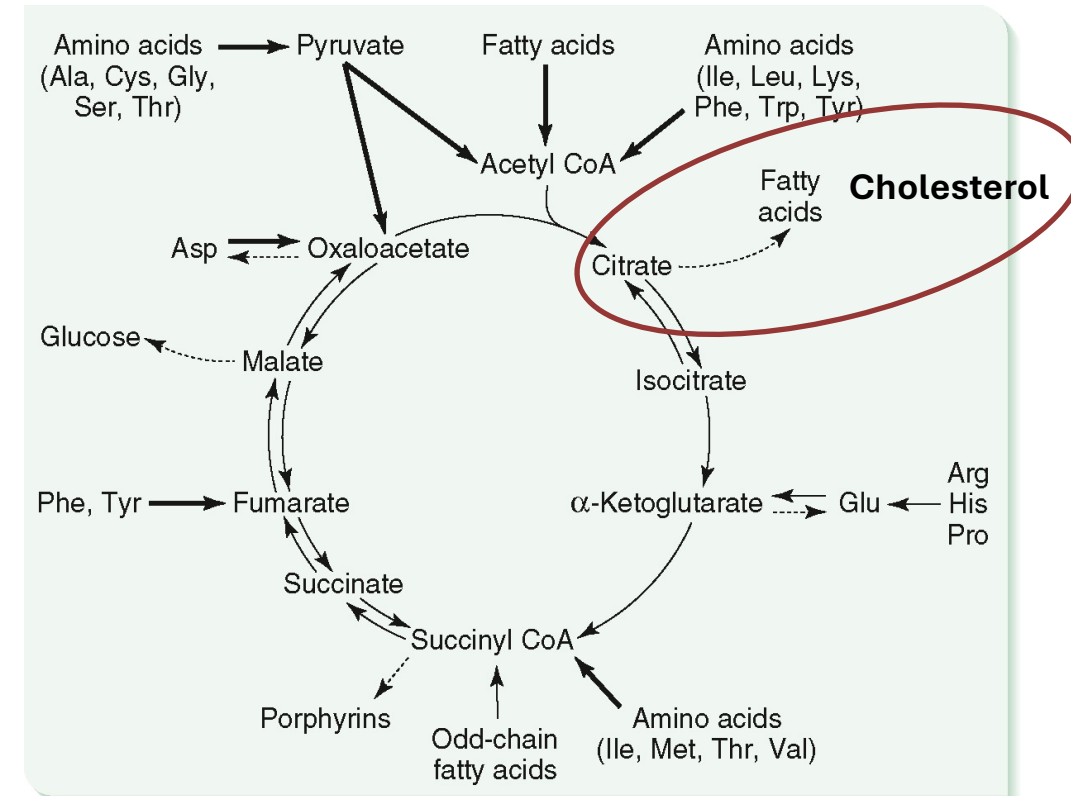
Intermediates of TCA Cycle

All intermediates of the TCA cycle are important for the generation of energy through the production of NADH, FADH₂, and GTP

Citrate, α -ketoglutarate, succinyl CoA and oxaloacetate also play key roles in other pathways

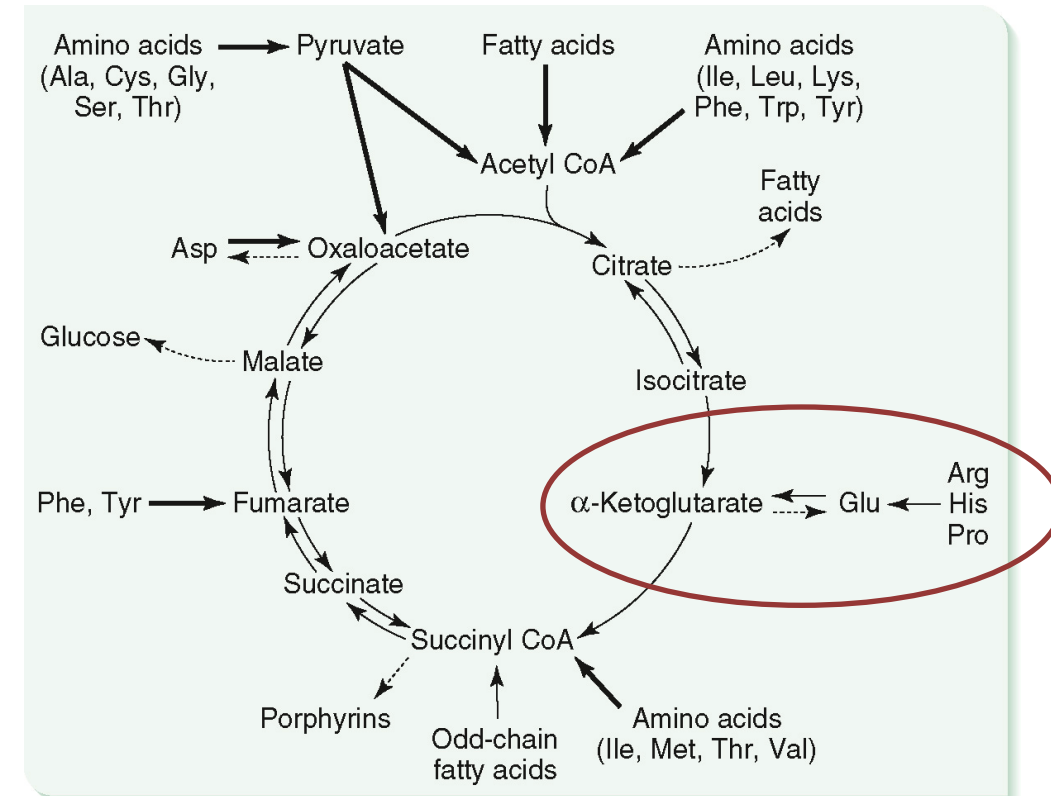
Physiological Importance of Citrate

- **Energy production:**
 - Oxidized through the TCA cycle to produce NADH and FADH₂ leading to generation of ATP through the electron transport chain
- **Regulation of metabolism:**
 - Allosteric inhibitor of phosphofructokinase-1 (glycolysis)
- **Biosynthesis of fatty acids:**
 - Citrate transported to cytosol
 - Used in synthesis of fatty acids and cholesterol
 - Allosteric activator of fatty acid synthesis



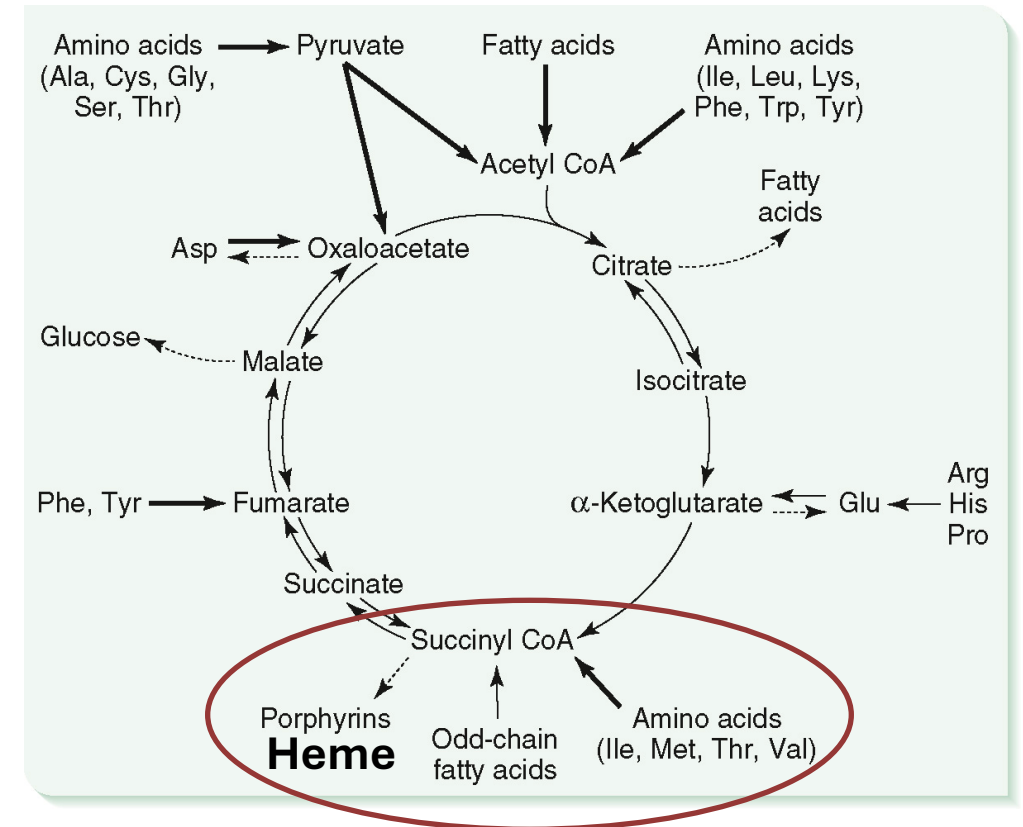
Physiological Importance of α -ketoglutarate

- **Energy production:**
 - oxidized through the TCA cycle to produce NADH and FADH₂ leading to generation of ATP through electron transport chain
- **Amino acid metabolism:**
 - Synthesis and degradation of amino acids (**glutamate**)
 - Substrate for aminotransferase enzymes; transfers amino group and forms glutamate
 - Conversion to glutamate
 - Precursor for other amino acids e.g. glutamine (precursor of purine/ pyridines); arginine; proline



Physiological Importance of Succinyl CoA

- **Energy production:**
 - oxidized through the TCA cycle to produce NADH and FADH₂ leading to generation of ATP through the electron transport chain
- **Heme synthesis:**
 - Precursor for the synthesis of heme
- **Amino acid metabolism:**
 - Synthesis and degradation of amino acids methionine, valine, isoleucine and threonine (Propionyl CoA → Succinyl CoA)
- **Fatty acid metabolism**
 - Can be formed from odd-chain fatty acids (Propionyl CoA → Succinyl CoA)

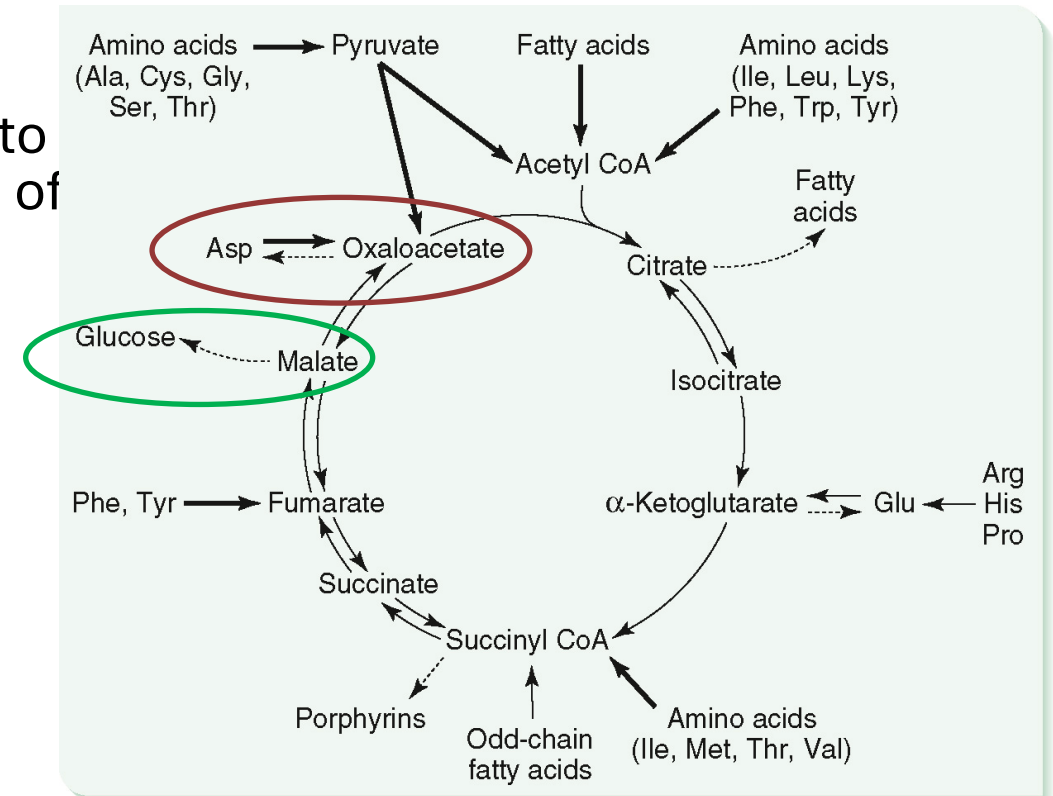


Physiological Importance of Oxaloacetate and malate

- **Oxaloacetate:**

- **Energy production:** Oxidized through TCA cycle to produce NADH and FADH₂ leading to generation of ATP through the electron transport chain
- **Gluconeogenesis:** Substrate for phosphoenolpyruvate carboxykinase (PEPCK)
- **Amino acid metabolism:**
 - Transamination reaction to form **aspartate** and α -ketoglutarate
 - Aspartate involved in *de novo* synthesis of purines and pyrimidines
 - Urea cycle (Aspartate is a substrate)

- **Malate** → Transported to cytosol → Gluconeogenesis



Malfunctions of Citric Acid Cycle

Fluoroacetate inhibits aconitase

substrate for citrate synthase, converted to fluorocitrate which then inhibits aconitase (binds to the Fe^{2+} ion at the active site)

Malonate inhibits Succinate Dehydrogenase

competitive inhibitor due to similarity of structure to succinate

Vitamin deficiencies:

Niacin (B_3)

Riboflavin (B_2)

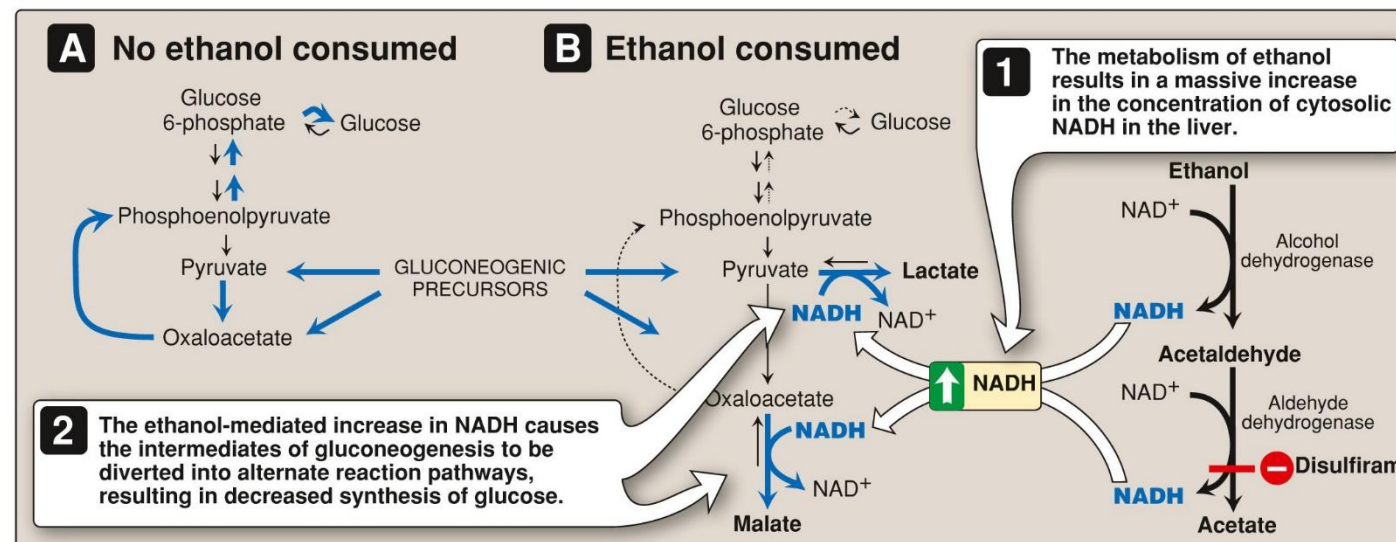
Thiamine (B_1)

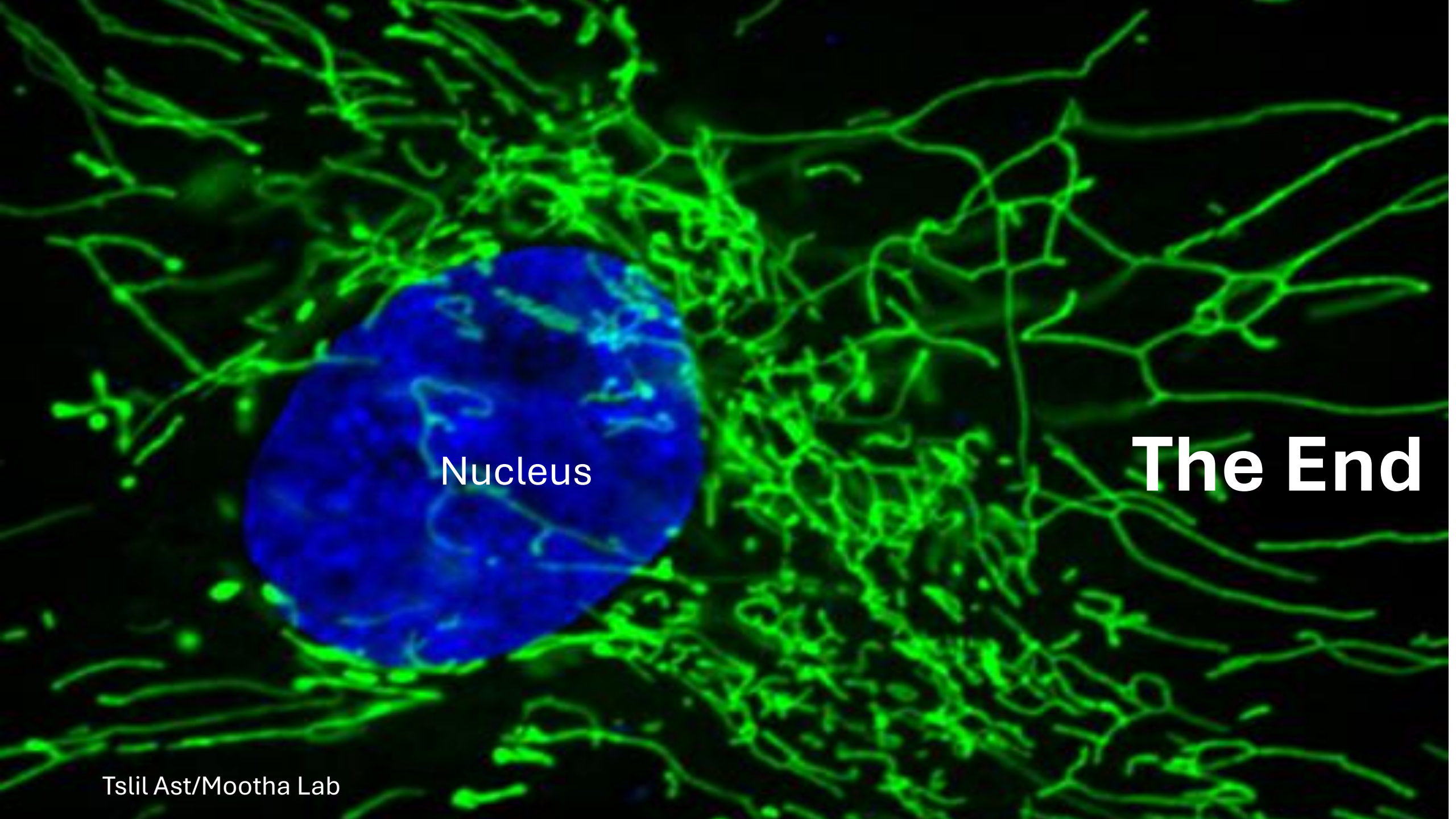
Pantothenate- B_5 (CoA is derived from B_5)

Cobalamin (B_{12}) (required for anaplerotic reaction which generates succinyl CoA from methylmalonyl CoA)

Binge alcohol consumption alters liver metabolism

- Increasing the NADH/NAD^+
- $\uparrow \text{NADH}/\text{NAD}^+$ alters reversible metabolic pathways:
 - Inhibition of PDH = $\downarrow$ [acetyl CoA] favoring pyruvate to lactate (due to $\uparrow \text{NADH}/\text{NAD}^+$)
 - **Lactic acidosis**
 - $\downarrow$ pyruvate = $\downarrow$ pyruvate carboxylase reaction (rate limiting step gluconeogenesis)
 - Inhibition of last steps of TCA cycle (malate to oxaloacetate reaction reverses)
 - **Malate trapped** further **inhibits gluconeogenesis**





A fluorescence microscopy image of a cell. The nucleus is stained blue and is located on the left side of the image. The cytoplasm and other cellular structures are stained green, showing a complex network of fibers and organelles. The background is black.

Nucleus

The End